TestLink

Test Specification

Test Project: UniShare

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2026 Group UniShare

Table Of Contents

1.Aland - Registration, Login & Admin Approval (Iteration 2)

1.1.TS-A1 Registration Form and Academic Selections

US-87: Public registration displays all mandatory controls

US-88: University list is alphabetical and faculties are scoped

US-89: Changing university clears the previously selected faculty

US-90: Each empty required registration input blocks submission

US-91: Remove uploaded student ID then replace it

1.2.TS-A2 Registration Input Validation

US-92: Name minimum boundaries at 1, 2 and 3 characters

US-93: Document maximum name boundary at 99, 100 and 101 characters

US-94: Email invalid partitions and valid nonuniversity domain

US-95: Existing approved and pending email cannot register twice

US-96: Password length boundaries at 7, 8 and 9 characters

US-97: Password complexity decision table for all three classes

US-98: Mismatched confirmation blocks then corrected value succeeds

US-99: API rejects empty academic selections and invalid ID URLs

US-100: Tampered faculty from a different university must be rejected

1.3.TS-A3 Student ID Upload Validation

US-101: Public image upload accepts JPG and PNG but rejects PDF

US-102: ID image size boundaries at 4 MiB minus one, exact and plus one byte

US-103: Two-file batch and failed upload do not enable registration

1.4.TS-A4 Registration Persistence and Email

US-104: Successful applicant is PENDING with hashed password and welcome email

US-105: SMTP welcome failure preserves a successful registration

US-106: Duplicate-email server error can be corrected without reuploading

1.5.TS-A5 Credential Login, Role Routing and Sessions

US-107: Approved student logs in with seeded credentials

US-108: Pending login cannot access student features and can sign out

US-109: Admin login reaches approval dashboard

US-110: Unknown email and wrong password have the same safe error

US-111: Missing and malformed login fields cannot authenticate

US-112: Existing sessions bypass login and signup by role

US-113: JWT session survives reload and another tab without exposing password

US-114: Pending session observes admin approval without signing in again

1.6.TS-A6 Admin Access and Applicant Review

US-115: Anonymous and nonadmin users cannot access admin pages or procedures

US-116: Pending list excludes approved/admin accounts, orders newest first and handles zero

US-117: Admin sees applicant identity and opens the stored student ID

US-118: Dashboard role counts refresh after approve and reject

US-119: Documented pending-applicant search and sorting are available

1.7.TS-A7 Approve Student Registration

US-120: Cancel approval leaves applicant pending

US-121: Confirm approval updates role, timestamp and email

US-122: Approve refuses already approved and admin targets

US-123: Approve rejects nonexistent or empty target ID

US-124: Approval remains effective when approval email fails

US-125: Sequential replay of approval preserves first approval timestamp

1.8.TS-A8 Reject Student Registration

US-126: Reject with reason sends explanation and deletes pending applicant

US-127: Optional rejection reason accepts blank and canceled prompt

US-128: Cancel final rejection confirmation preserves applicant

US-129: Reject refuses approved/admin targets and missing users

US-130: Rejection still deletes applicant when mail fails

US-131: Deleted applicant cannot login and may register again for review

2.Karoz - Course, Resource & Timetable Management (Iteration 3 UC1, Iteration 4 UC-S4-005)

2.1.TS-K1 Create Course (UC-S3-001)

US-132: Create a course with title, description and colour (main flow)

US-133: Title length boundaries (0, 1, 100, 101 characters)

US-134: Whitespace-only and padded title

US-135: Description length boundaries (empty, 500, 501 characters)

US-136: Colour selection - each swatch

US-137: Colour format validation in the API

US-138: Four default resource cards are created

US-139: Duplicate course titles are allowed

US-140: Course creation blocked for visitors and PENDING students

US-141: course.create called without a session or with a PENDING session

2.2.TS-K2 Manage Courses

US-142: Course list shows owned and shared courses only

US-143: Course access by role

US-144: Owner edits title, description and colour

US-145: Non-owner cannot edit or open settings

US-146: Clearing the description when editing a course

US-147: Delete course - confirm and cancel

US-148: Non-owner cannot delete a course

US-149: Toggle favourite on and off

2.3.TS-K3 Resources & Files

US-150: Owner creates a custom resource card

US-151: Resource title and description boundaries

US-152: Deadline states: future, within 3 days, overdue

US-153: Create/update permission matrix

US-154: Delete resource - owner only, custom cards only in the UI

US-155: Attach files to a card that allows uploads

US-156: File upload refused for cards without uploads and invalid URLs

US-157: Upload limits: size, type and count

US-158: Remove a file from a card

2.4.TS-K4 Timetable (UC-S4-005)

US-159: Create a timetable (first-time and name boundaries)

US-160: Set, keep and lose the default timetable

US-161: Add a weekly class to the timetable (main flow)

US-162: Add Class with no favourite courses

US-163: API refuses classes for non-favourite courses

US-164: Class title and location length boundaries

US-165: End time must be after start time

US-166: Time conflict detection boundaries

US-167: Maximum 3 classes per course per day

US-168: Edit a class in the details dialog

US-169: Drag and resize a class in the calendar

US-170: Delete a class

US-171: Timetable role matrix (owner, contributor, viewer, outsider)

US-172: Invite a student to a timetable (search length, role, duplicates)

US-173: Timetable invitation lifecycle: accept, reject, leave

US-174: Class reminder is created at the configured time

US-175: Rename and delete a timetable (owner only)

3.Parwar - AI Assistant, Quizzes, Study Plans & Collaborative Notes (Iterations 3-4)

3.1.TS-P1 Chat with AI Assistant (UC-S3-012)

US-42: Start a new AI conversation (main flow)

US-43: Empty and whitespace-only message is not sent

US-44: Message length boundaries (4999, 5000, 5001 characters)

US-45: Continue a previous conversation with context

US-46: Chat access without approval, session or conversation id

US-47: AI server unavailable

3.2.TS-P2 Conversation History & Privacy (UC-S3-019/020)

US-48: Sidebar paging boundaries (5, 6 and 16 conversations)

US-49: New Chat and untitled conversation

US-50: Delete conversation - confirm and cancel

US-51: User cannot read, continue or delete another user's conversation

3.3.TS-P3 Generate AI Quiz (UC-S3-014)

US-52: Generate a quiz for own course (main flow)

US-53: Topic is required

US-54: No question type selected

US-55: Question count boundaries

US-56: Difficulty and question type partitions

US-57: Quiz generation access by course relation

US-58: Invalid AI response is not saved

3.4.TS-P4 Attempt Quiz & Grading (UC-S3-015)

US-59: Submit is enabled only when all questions are answered

US-60: Score bands on the results page

US-61: Answer matching rules per question type

US-62: Attempt is saved, best score and retake

US-63: Another user's quiz cannot be opened, attempted or deleted

US-64: Delete own quiz

3.5.TS-P5 AI Study Plans (UC-S3-016/017)

US-65: Generate a study plan (main flow)

US-66: Weeks and hours boundaries

US-67: Course is required and deadline cannot be in the past

US-68: Study plan access by course relation

US-69: Track task completion and progress

US-70: Plan ownership and delete

US-71: Study plan survives course deletion

3.6.TS-P6 Enhance Notes with AI (UC-S3-018)

US-72: Each enhancement type returns content and is recorded

US-73: Invalid enhancement requests

US-74: Enhancement permission by role

3.7.TS-P7 Collaborative Notes - Access & Roles (UC-S4-002)

US-75: Owner opens notes for the first time

US-76: Viewer opens a course without pages

US-77: Notes access by user relation

US-78: Viewer is read-only (UI and server)

US-79: Liveblocks authentication endpoint

US-80: Unknown or other course page id

3.8.TS-P8 Collaborative Notes - Real-Time Editing & Pages

US-81: Two students edit the same page in real time

US-82: Simultaneous edit of the same text converges

US-83: Presence list boundaries and leaving

US-84: Create page and nested sub-page

US-85: Delete a page with sub-pages

US-86: Offline edits sync after reconnect

4.Rawa Dara - Sharing, Articles & Notifications (Iteration 4)

4.1.TS-R1 Share Course (UC-S4-001)

US-1: Owner invites same-faculty student as VIEWER

US-2: Owner invites same-faculty student as CONTRIBUTOR

US-3: Non-owner cannot invite collaborators

US-4: Owner cannot invite himself

US-5: Duplicate invitation to the same student is rejected

US-6: Invite to a course that does not exist

US-7: Invitation to a student from a different faculty is blocked

US-8: Owner changes collaborator permission and removes collaborator

4.2.TS-R2 Respond to Course Invitation

US-9: Invitee accepts a pending course invitation

US-10: Invitee declines a pending course invitation

US-11: Answered invitation cannot be accepted or declined again

4.3.TS-R3 Shared Course Permissions (Viewer / Contributor)

US-12: VIEWER can view and download but cannot modify

US-13: CONTRIBUTOR can add resources and upload files

US-14: CONTRIBUTOR cannot delete resources or the course

US-15: Non-collaborator and declined user cannot open a private course

4.4.TS-R4 Create & Edit Article (UC-S4-003)

US-16: Create article saved as draft (valid data)

US-17: Title length boundaries (0, 1, 200, 201 characters)

US-18: Excerpt length boundaries (500 / 501 characters)

US-19: Cover image URL validation

US-20: Duplicate titles get unique slugs

US-21: Author edits own article; tags are replaced

US-22: User cannot edit another user's article

4.5.TS-R5 Article Lifecycle (Publish / Archive / Delete)

US-23: Publish a draft article

US-24: Only DRAFT articles can be published

US-25: Archive own published article

US-26: Delete own article

US-27: Non-author cannot publish, archive or delete

4.6.TS-R6 Browse & Read Public Articles

US-28: Access control by status and viewer

US-29: Listing shows only published, newest first, with pagination

US-30: Search and tag filter

US-31: View counter increments only for published articles

US-32: My Articles status filter

4.7.TS-R7 Notification Centre (UC-S4-004)

US-33: Unread badge boundaries (0, 1, 9, 10)

US-34: Dropdown lists 20 most recent, newest first

US-35: Click invitation notification marks read and opens Pending Invitations

US-36: Mark all as read

US-37: User cannot read or mark another user's notification

US-38: View all notifications page with paging

4.8.TS-R8 Notification Preferences

US-39: Default preferences are created on first visit

US-40: Save in-app / email preferences per type

US-41: Class reminder timing boundaries

Scope

UniShare - university course sharing platform (Next.js / tRPC / Prisma). Group project for SECJ3343 Project 1 (Aland, Karoz, Parwar, Rawa). Source: github.com/Seganation/unishare

1.Test Suite : Aland - Registration, Login & Admin Approval (Iteration 2)

Tester: Aland (GitHub Aland95)
Software under test: UniShare main, authentication implementation b124314.
Test basis: assignment/UniShare Iteration 2.md detailed Use Cases 1-3: UC-S2-001 Student Registration, UC-S2-002 Admin Approval, UC-S2-003 Student Login; src/server/api/routers/user.ts, admin.ts, src/server/auth/config.ts, middleware.ts, src/app/(auth) and src/app/(admin), src/app/(student)/layout.tsx, src/app/api/uploadthing/core.ts, src/lib/email.ts, prisma/schema.prisma.
Techniques: Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table, State Transition, Use Case, Error Guessing.

Common test data: From prisma/seed.ts: Admin User, admin@unishare.com (ADMIN); Sarah Ahmed, sarah@student.utm.my (APPROVED); John Doe, john@student.utm.my (APPROVED); Alice Wong, alice@student.utm.my (PENDING); Michael Chen, michael@student.utm.my (APPROVED, Faculty of Engineering). All seeded passwords are admin123. Admin, Sarah, John and Alice use Universiti Teknologi Malaysia / Faculty of Computing. Seed passwords predate the new-registration uppercase rule; login does not apply registration password complexity.

Execution setup: UniShare runs on http://localhost:3000 with a disposable seeded Postgres database, configured UploadThing and a controlled SMTP capture/stub. Restore the initial fixture before each independent case; never reseed the group's TestLink database. The new-applicant baseline is name 'Aland QA', unused email aland.qa.01@example.com, password and confirmation 'Test1234', seeded UTM and Computing IDs obtained from user.getUniversities, and a successfully uploaded student-id.jpg image of 1 MiB. API baseline studentIdUrl is https://utfs.io/f/aland-qa-id.jpg; it is a validation fixture, not a promised live image. Use a fresh aland.qa.N@example.com email for every accepted registration within a case. API steps use a local tRPC client/server caller with the indicated session and exact JSON input; inspect responses in developer tools. Database checks use Prisma Studio or a local query. File sizes are exact bytes; 1 MiB = 1048576 bytes.

Oracle policy: Current code supplies URLs, limits and exact application messages. Requirements explicitly labelled document-only retain the iteration's business expectation and identify the missing check; do not mark them passed from code inspection. Browser-native validation and UploadThing errors vary by version; assert validity/error state and the application's literal 'Upload failed: ' prefix, rather than inventing library text. Numbered steps are generated by load.py. All cases remain Not Run for Project 1.

1.1.Test Suite : TS-A1 Registration Form and Academic Selections

UC-S2-001 form paths, cascading selections, required inputs and ID replacement. EP, Use Case, State Transition.

Test Case US-87: Public registration displays all mandatory controls [Version : 1]		
Author:	aland	
Summary: Public registration displays all mandatory controls. **Test design technique:** Use Case (registration entry), EP (anonymous visitor) **Requirements:** REQ-A-REG-01, REQ-A-REG-10		
Preconditions: Fresh anonymous browser; seeded universities available; no file uploaded.		
#:	Step actions:	Expected Results:
1	Open /login and click 'Sign Up Here'.	URL is /signup; heading 'Join UNIShare' is displayed.
2	Inspect all form controls.	Labels 'Full Name', 'University Email', 'Password', 'Confirm Password', 'University', 'Faculty', 'Student ID Photo' are displayed; password fields are masked.
3	Inspect 'Create Account' before uploading.	'Create Account' is disabled; no registration request or user record is created.
Execution type:	Manual	

Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-01: Public registration form and required fields [Version : 1] REQ-A-REG-10: Required uploaded ID and URL validation [Version : 1]
Keywords:	None

Test Case US-88: University list is alphabetical and faculties are scoped [Version : 1]		
Author:	aland	
Summary: University list is alphabetical and faculties are scoped. **Test design technique:** EP (unselected/selected university), Use Case **Requirements:** REQ-A-REG-02		
Preconditions: Anonymous /signup; original seed contains Universiti Malaya, Universiti Sains Malaysia, Universiti Teknologi Malaysia.		
#:	Step actions:	Expected Results:
1	Inspect university and faculty before selection.	Placeholders 'Select university' and 'Select faculty'; Faculty is disabled.
2	Open University and select Universiti Teknologi Malaysia; inspect user.getUniversities response.	Universities ascend by name: Malaya, Sains Malaysia, Teknologi Malaysia; UTM faculty options ascend Computing, Education, Engineering, Management, Science.
3	Select Faculty of Computing.	Selected faculty is 'Faculty of Computing'; no Malaya-specific Faculty of Computer Science and Information Technology option appears.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-REG-02: University and faculty selection [Version : 1]	
Keywords:	None	

Test Case US-89: Changing university clears the previously selected faculty [Version : 1]		
Author:	aland	
Summary: Changing university clears the previously selected faculty. **Test design technique:** State Transition (UTM selected -> Malaya selected), Error Guessing **Requirements:** REQ-A-REG-02, REQ-A-REG-11		
Preconditions: Anonymous /signup, baseline valid text fields and uploaded ID; UTM / Faculty of Computing selected.		
#:	Step actions:	Expected Results:
1	Change University to Universiti Malaya.	Faculty resets to 'Select faculty'; UTM Faculty of Computing is no longer an option.
2	Click 'Create Account' without selecting a new faculty.	Native required-field validation blocks submission at 'Faculty'; no user.register request is sent.

3	Select Faculty of Computer Science and Information Technology.	New faculty belongs to Malaya; required-field validity is restored.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-REG-02: University and faculty selection [Version : 1] REQ-A-REG-11: Required academic selections [Version : 1]	
Keywords:	None	

Test Case US-90: Each empty required registration input blocks submission [Version : 1]		
Author:	aland	
Summary: Each empty required registration input blocks submission. **Test design technique:** Decision Table (one missing mandatory input per row), EP negative **Requirements:** REQ-A-REG-01, REQ-A-REG-11		
Preconditions: Anonymous /signup with valid baseline and uploaded ID; restore baseline before each row; inspect Network panel.		
#:	Step actions:	Expected Results:
1	For each row separately, clear Full Name, University Email, Password, or Confirm Password and click 'Create Account'.	Each empty required field fails native validity; no user.register request and no applicant is created; exact browser bubble wording is browser-dependent.
2	Restore baseline; set University back to 'Select university' and submit.	University is invalid, Faculty is cleared; no registration request.
3	Restore baseline; set Faculty to 'Select faculty' and submit.	Faculty is invalid; no registration request; 'Create Account' remains available after correction.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-REG-01: Public registration form and required fields [Version : 1] REQ-A-REG-11: Required academic selections [Version : 1]	
Keywords:	None	

Test Case US-91: Remove uploaded student ID then replace it [Version : 1]		
Author:	aland	
Summary: Remove uploaded student ID then replace it. **Test design technique:** State Transition (uploaded -> absent -> uploaded) **Requirements:** REQ-A-REG-10, REQ-A-REG-15		
Preconditions: Anonymous /signup with baseline fields and a successful 1 MiB student-id.jpg upload; keep returned first URL.		
#:	Step actions:	Expected Results:
1	Check completed upload and click 'Remove'.	'Student ID uploaded' disappears; uploader returns; 'Create Account' is disabled.

2	Upload replacement-id.png (valid 1 MiB PNG).	'Student ID uploaded' and 'Remove' reappear; 'Create Account' is enabled.
3	Submit and inspect user.register payload and stored row.	Payload/studentIdUrl use replacement URL, not old URL; UI displays 'Registration Successful!'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-A-REG-10: Required uploaded ID and URL validation [Version : 1] REQ-A-REG-15: Upload removal and recovery [Version : 1]	
Keywords:	None	

1.2.Test Suite : TS-A2 Registration Input Validation

UC-S2-001 EP, BVA, Decision Table and tampering. Business-rule gaps remain explicit expected failures.

Test Case US-92: Name minimum boundaries at 1, 2 and 3 characters [Version : 1]		
Author:	aland	
Summary: Name minimum boundaries at 1, 2 and 3 characters. **Test design technique:** BVA (min-1/min/min+1 = 1/2/3) **Requirements:** REQ-A-REG-03		
Preconditions: Public user.register client; baseline valid payload; unused aland.qa.06a/b/c@example.com per row.		
#:	Step actions:	Expected Results:
1	Call user.register with name 'A' (1 character).	Validation issue 'Name must be at least 2 characters'; no user is inserted.
2	Repeat with name 'Al' (2), fresh email.	success=true; message 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'; name stored as Al.
3	Repeat with name 'Ali' (3), fresh email.	success=true with the same exact success message; name stored as Ali.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-REG-03: Minimum name length [Version : 1]	
Keywords:	None	

Test Case US-93: Document maximum name boundary at 99, 100 and 101 characters [Version : 1]		
Author:	aland	
Summary: Document maximum name boundary at 99, 100 and 101 characters. **Test design technique:** BVA (max-1/max/max+1 = 99/100/101), Error Guessing **Requirements:** REQ-A-REG-04		
Preconditions:		

Public user.register with valid baseline; fresh aland.qa.07a/b/c@example.com per input. Document-derived case; current code lacks max(100).

#:	Step actions:	Expected Results:
1	Register with name exactly 'A' repeated 99 times and a fresh email.	Accepted: success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'.
2	Register with name exactly 'A' repeated 100 times and a fresh email.	Accepted: same success message; full 100-character name stored.
3	Register with name exactly 'A' repeated 101 times and a fresh email.	Document expectation: validation rejection, no new user. Current implementation instead accepts it; record discrepancy against the 100-character rule. No app maximum-name error text exists.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-A-REG-04: Maximum name length of 100 (document-only gap) [Version : 1]
Keywords:	None

Test Case US-94: Email invalid partitions and valid nonuniversity domain [Version : 1]

Author:	aland

Summary:

Email invalid partitions and valid nonuniversity domain.

Test design technique: EP (empty/missing-at/missing-domain/valid)

Requirements: REQ-A-REG-05

Preconditions:

Public user.register; baseline except email; aland.qa.08@example.com unused. Anonymous /signup available for UI check.

#:	Step actions:	Expected Results:
1	Individually submit email '', 'aland.example.com', and 'aland@' through API.	Every input returns validation issue 'Invalid email address'; no applicants created.
2	Set UI University Email to 'aland.example.com' with other fields valid; submit.	Native email validity fails; user.register is not called; browser wording is not app-controlled.
3	Register through API using aland.qa.08@example.com.	Accepted: success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'; no university-domain restriction is applied.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-05: Email syntax [Version : 1]
Keywords:	None

Test Case US-95: Existing approved and pending email cannot register twice [Version : 1]

Author:	aland

Summary:

Existing approved and pending email cannot register twice.

Test design technique: EP (duplicate APPROVED/PENDING), Error Guessing

Requirements: REQ-A-REG-06

Preconditions:

Original seed: Sarah APPROVED and Alice PENDING; baseline password Test1234 meets registration rules; capture their initial rows.

#:	Step actions:	Expected Results:
1	Call user.register with sarah@student.utm.my and otherwise valid baseline.	Error 'User with this email already exists'; Sarah's role, password hash and details unchanged.
2	Repeat with alice@student.utm.my.	Same error 'User with this email already exists'; one Alice row remains PENDING with original details.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-06: Unique registration email and error recovery [Version : 1]
Keywords:	None

Test Case US-96: Password length boundaries at 7, 8 and 9 characters [Version : 1]

Author:	aland

Summary:

Password length boundaries at 7, 8 and 9 characters.

Test design technique: BVA (min-1/min/min+1 = 7/8/9)

Requirements: REQ-A-REG-07

Preconditions:

Anonymous /signup with baseline including uploaded ID; reset form between accepted rows, fresh aland.qa.10a/b@example.com.

#:	Step actions:	Expected Results:
1	Set password and confirmation to 'Test123' (7), click 'Create Account'.	'Password must be at least 8 characters'; no registration request; form reenabled.
2	Set both to 'Test1234' (8), submit with fresh email.	'Registration Successful!'; applicant created PENDING.
3	Return to /signup; fill baseline with fresh email and both passwords 'Test12345' (9); submit.	'Registration Successful!'; 9-character password is accepted.
4	Call API user.register with 'Test123' (7) in otherwise valid baseline.	Server independently rejects with 'Password must be at least 8 characters'; no user inserted.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-07: Password minimum length [Version : 1]
Keywords:	None

Test Case US-97: Password complexity decision table for all three classes [Version : 1]

Author:	aland

Summary:

Password complexity decision table for all three classes.

Test design technique: Decision Table (uppercase/lowercase/digit), EP

Requirements: REQ-A-REG-08

Preconditions:

Public user.register; baseline valid fields; distinct fresh email for each accepted row.

#:	Step actions:	Expected Results:
1	Submit 'test1234' (no uppercase), 'TEST1234' (no lowercase), 'Testabcd' (no digit), each separately.	Each rejected with 'Password must contain at least one uppercase letter, one lowercase letter, and one number'; no applicant created.
2	Submit '!!!!!!!!' (none of the three required classes).	Same exact complexity error; no applicant created.
3	Submit 'Test1234' (all classes, no symbol) then 'Test123!' (all classes plus symbol), with fresh emails.	Both accepted with success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-08: Password character classes [Version : 1]
Keywords:	None

Test Case US-98: Mismatched confirmation blocks then corrected value succeeds [Version : 1]

Author:	aland

Summary:

Mismatched confirmation blocks then corrected value succeeds.

Test design technique: EP (unequal/equal), State Transition (error -> corrected)

Requirements: REQ-A-REG-09

Preconditions:

Anonymous /signup with baseline and uploaded ID, unused aland.qa.12@example.com; Network panel open.

#:	Step actions:	Expected Results:
1	Enter password 'Test1234', confirmation 'Test1235'; click 'Create Account'.	Exact error 'Passwords do not match'; no user.register request; fields become editable.
2	Change confirmation to 'Test1234' and submit again.	Error cleared; one request succeeds; 'Registration Successful!' displayed; exactly one applicant created.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-09: Password confirmation [Version : 1]
Keywords:	None

Test Case US-99: API rejects empty academic selections and invalid ID URLs [Version : 1]

Author:	aland

Summary:

API rejects empty academic selections and invalid ID URLs.

Test design technique: EP (missing academic value / malformed URL / valid URL), Error Guessing

Requirements: REQ-A-REG-10, REQ-A-REG-11

Preconditions:

Public user.register; valid baseline except the field under test. Restore baseline per row; no applicant for rejected input.

#:	Step actions:	Expected Results:
1	Submit universityId='' then restore and submit facultyId=''.	Respectively 'University is required' and 'Faculty is required'; no user created.
2	Submit studentIdUrl='' and then 'not-a-url', separately.	Both return 'Invalid student ID URL'; no user created.
3	Submit studentIdUrl='https://example.com/id.png', all other fields valid, fresh email.	API accepts syntactically valid URL, success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'. Current schema has no UploadThing-host allowlist.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-REG-10: Required uploaded ID and URL validation [Version : 1] REQ-A-REG-11: Required academic selections [Version : 1]	
Keywords:	None	

Test Case US-100: Tampered faculty from a different university must be rejected [Version : 1]

Author:	aland

Summary:

Tampered faculty from a different university must be rejected.

Test design technique: Decision Table (selected university matches faculty owner / does not match), Error Guessing

Requirements: REQ-A-REG-12

Preconditions:

Fetch real IDs with user.getUniversities. U=UTM, F=Universiti Malaya Faculty of Computer Science and Information Technology; baseline valid, unused aland.qa.14@example.com. Document-derived negative case.

#:	Step actions:	Expected Results:
1	Call user.register with universityId=U and facultyId=F (both exist, but F belongs to Malaya).	Document expectation: reject with 'Invalid faculty selection' and create no applicant. Current code has no relationship check and is expected to accept this pair; record the discrepancy, not a passing result.
2	Repeat with UTM's actual Computing faculty ID and a different unused email.	Valid pair accepted with 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-REG-12: Faculty must belong to selected university (document-only gap) [Version : 1]	
Keywords:	None	

1.3.Test Suite : TS-A3 Student ID Upload Validation

UC-S2-001 public image upload: type, exact file-size boundary, file count and failure recovery. EP, BVA, Error Guessing.

Test Case US-101: Public image upload accepts JPG and PNG but rejects PDF [Version : 1]		
Author:	aland	
Summary: Public image upload accepts JPG and PNG but rejects PDF. **Test design technique:** EP (supported image / unsupported document), Use Case **Requirements:** REQ-A-REG-13		
Preconditions: Anonymous /signup; working UploadThing; genuine valid student-id.jpg, student-id.png and student-id.pdf, each 1 MiB; no session.		
#:	Step actions:	Expected Results:
1	Upload student-id.jpg through studentIdUploader; inspect completion response.	Upload succeeds without login; response uploadedBy='registration' and fileUrl is returned; 'Student ID uploaded' displayed.
2	Click 'Remove'; upload student-id.png.	PNG accepted; 'Student ID uploaded' displayed; one active URL selected.
3	Click 'Remove'; attempt student-id.pdf.	File selection or uploader validation rejects PDF; 'Student ID uploaded' is absent and 'Create Account' disabled. Any application callback error begins 'Upload failed: '; library suffix is version-dependent.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-REG-13: Public upload accepts one image [Version : 1]	
Keywords:	None	

Test Case US-102: ID image size boundaries at 4 MiB minus one, exact and plus one byte [Version : 1]		
Author:	aland	
Summary: ID image size boundaries at 4 MiB minus one, exact and plus one byte. **Test design technique:** BVA (4194303 / 4194304 / 4194305 bytes) **Requirements:** REQ-A-REG-14		
Preconditions: Anonymous /signup; valid decodable PNG fixtures id-below.png=4194303 bytes, id-exact.png=4194304, id-above.png=4194305; inspect sizes on disk; working UploadThing.		
#:	Step actions:	Expected Results:
1	Upload id-below.png (4194303 bytes).	Accepted; 'Student ID uploaded'; 'Create Account' enabled when upload completes.
2	Remove; upload id-exact.png (4194304 bytes).	Accepted at inclusive maximum; 'Student ID uploaded'.
3	Remove; attempt id-above.png (4194305 bytes).	Rejected over maximum; no selected URL; 'Create Account' disabled. If callback runs, application error prefix is 'Upload failed: '; do not substitute the iteration's unimplemented custom message.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	

Requirements	REQ-A-REG-14: Student ID file size maximum [Version : 1]
Keywords:	None

Test Case US-103: Two-file batch and failed upload do not enable registration [Version : 1]		
Author:	aland	
Summary: Two-file batch and failed upload do not enable registration. **Test design technique:** BVA (file count 1/2), Error Guessing (network failure), State Transition **Requirements:** REQ-A-REG-13, REQ-A-REG-15		
Preconditions: Anonymous /signup with baseline valid fields; two genuine 1 MiB images; developer-tools request interception or upload test stub available.		
#:	Step actions:	Expected Results:
1	Submit a single upload batch containing both images through studentIdUploader (use API if picker limits selection).	Batch rejected by maxFileCount=1; no successful completed two-image upload. 'Create Account' remains disabled until one upload succeeds.
2	Attempt one image and inject uploader onUploadError with Error('QA upload unavailable').	Exact UI error 'Upload failed: QA upload unavailable'; no 'Student ID uploaded'; 'Create Account' disabled.
3	Remove fault injection and retry one image.	Upload completes; 'Student ID uploaded' and 'Remove' displayed; 'Create Account' enabled.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-A-REG-13: Public upload accepts one image [Version : 1] REQ-A-REG-15: Upload removal and recovery [Version : 1]	
Keywords:	None	

1.4.Test Suite : TS-A4 Registration Persistence and Email

UC-S2-001 end-to-end submission, bcrypt, PENDING state, email delivery failure and form recovery. Use Case, State Transition, Error Guessing.

Test Case US-104: Successful applicant is PENDING with hashed password and welcome email [Version : 1]		
Author:	aland	
Summary: Successful applicant is PENDING with hashed password and welcome email. **Test design technique:** Use Case (registration main flow), State Transition (absent -> PENDING) **Requirements:** REQ-A-REG-16, REQ-A-REG-17		
Preconditions: Anonymous /signup; baseline with unused aland.qa.18@example.com and real uploaded image; controlled SMTP capture succeeds; disposable database inspection available.		
#:	Step actions:	Expected Results:
1	Complete baseline fields; click 'Create Account'; capture mutation request/response.	During submission fields disable and button says 'Creating...'; response success=true with userId and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'.

2	Observe immediate success panel and wait at least 3 seconds.	'Registration Successful!', 'Your account has been created and is pending admin approval.', 'Redirecting to login...' displayed, then /login; no authenticated applicant session is automatically created.
3	Inspect the new User and verify bcrypt.compare('Test1234', stored password) in a local verification tool.	Exactly one user: name Aland QA, role PENDING, approvedAt null, selected UTM/Computing IDs and returned ID URL; timestamps set; password differs from plaintext, bcrypt cost 12, comparison true.
4	Inspect captured welcome mail addressed to aland.qa.18@example.com.	Subject 'Welcome to UNIShare - Registration Received'; body contains 'Your account status:' and 'PENDING'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1] REQ-A-REG-17: Welcome email is best effort [Version : 1]
Keywords:	None

Test Case US-105: SMTP welcome failure preserves a successful registration [Version : 1]

Author:	aland

Summary:

SMTP welcome failure preserves a successful registration.

Test design technique: Error Guessing (SMTP failure), State Transition

Requirements: REQ-A-REG-16, REQ-A-REG-17, REQ-A-REG-06

Preconditions:

Public user.register with baseline aland.qa.19@example.com unused; controlled mail transport throws Error('QA SMTP down'); restore transport afterward.

#:	Step actions:	Expected Results:
1	Register baseline applicant while SMTP fails.	success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'; server logs 'Failed to send welcome email:'.
2	Inspect database, then retry same registration email.	One PENDING applicant with null approvedAt remains; retry returns 'User with this email already exists' and does not insert a second user.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-06: Unique registration email and error recovery [Version : 1] REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1] REQ-A-REG-17: Welcome email is best effort [Version : 1]
Keywords:	None

Test Case US-106: Duplicate-email server error can be corrected without reuploading [Version : 1]

Author:	aland

Summary:

Duplicate-email server error can be corrected without reuploading.

Test design technique: State Transition (submitting -> error -> success), Error Guessing

Requirements: REQ-A-REG-06, REQ-A-REG-16

Preconditions:		
Anonymous /signup; valid baseline and uploaded ID; email initially sarah@student.utm.my; aland.qa.20@example.com unused.		
#:	Step actions:	Expected Results:
1	Submit existing Sarah email with otherwise valid Test1234 baseline.	Exact error 'User with this email already exists'; loading clears, 'Create Account' and fields reenabled; selected 'Student ID uploaded' remains.
2	Change only email to aland.qa.20@example.com and submit.	Old error clears; 'Registration Successful!'; one new PENDING applicant, original Sarah unchanged.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-A-REG-06: Unique registration email and error recovery [Version : 1] REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1]	
Keywords:	None	

1.5.Test Suite : TS-A5 Credential Login, Role Routing and Sessions

UC-S2-003 and Admin Login: role decision table, bad credentials, session persistence, fresh role and sign out. EP, Decision Table, State Transition, Use Case.

Test Case US-107: Approved student logs in with seeded credentials [Version : 1]		
Author:	aland	
Summary:		
Approved student logs in with seeded credentials. **Test design technique:** Use Case (login main flow), Decision Table (APPROVED + correct password) **Requirements:** REQ-A-LOG-01		
Preconditions:		
Anonymous fresh browser; seed Sarah APPROVED; password admin123; UniShare and database available.		
#:	Step actions:	Expected Results:
1	Open /login; enter sarah@student.utm.my and admin123; click 'Sign In'.	'Signing In...' then successful overlay 'Welcome Back!' and 'Taking you to your dashboard...'.
2	Wait for redirect and inspect current session.	Final URL /courses; 'Courses' navigation visible; session user is Sarah with role APPROVED. Seeded lowercase admin123 is accepted despite new-registration uppercase rule.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-LOG-01: Valid credentials route by current role [Version : 1]	
Keywords:	None	

Test Case US-108: Pending login cannot access student features and can sign out [Version : 1]	
Author:	aland
Summary:	
Pending login cannot access student features and can sign out. **Test design technique:** Decision Table (PENDING + correct password), State Transition (authenticated -> signed out)	

Requirements: REQ-A-LOG-01, REQ-A-LOG-06

Preconditions:

Fresh browser; original seed Alice PENDING with null approvedAt; password admin123.

#:	Step actions:	Expected Results:
1	Sign in on /login with alice@student.utm.my / admin123.	Login succeeds, final URL /waiting-approval, heading 'Account Pending Approval'; text 'Your registration is currently under review by our admin team.'.
2	Individually open /courses, /timetable, /my-articles and /settings.	Each finishes at /waiting-approval; 'Account Pending Approval' displayed instead of student content.
3	Click 'Sign Out'; query user.getCurrentSession and revisit /courses.	Returns /login; getCurrentSession is null; /courses redirects /login with 'Sign In' button.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-LOG-01: Valid credentials route by current role [Version : 1] REQ-A-LOG-06: Pending access restrictions and sign out [Version : 1]
Keywords:	None

Test Case US-109: Admin login reaches approval dashboard [Version : 1]

Author:	aland

Summary:

Admin login reaches approval dashboard.

Test design technique: Decision Table (ADMIN + correct password), Use Case (Admin Login)

Requirements: REQ-A-LOG-01, REQ-A-ADM-01

Preconditions:

Anonymous browser; Admin User admin@unishare.com / admin123 has ADMIN role.

#:	Step actions:	Expected Results:
1	Sign in with admin@unishare.com / admin123.	'Welcome Back!' then redirects through /courses to /admin/approvals; 'UNIShare Admin' and 'Student Approvals' visible.
2	Open /admin and then /courses directly.	Both finish at /admin/approvals; 'Student Approvals' visible; admin is not left on the student page.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-01: Admin route and API authorization [Version : 1] REQ-A-LOG-01: Valid credentials route by current role [Version : 1]
Keywords:	None

Test Case US-110: Unknown email and wrong password have the same safe error [Version : 1]

Author:	aland

Summary:

Unknown email and wrong password have the same safe error.

Test design technique: EP (unknown account / bad password), Decision Table negative

Requirements: REQ-A-LOG-02

Preconditions:

Anonymous /login; missing.qa@example.com absent; Sarah exists; query user.getCurrentSession after each failure.

#:	Step actions:	Expected Results:
1	Submit missing.qa@example.com / admin123.	UI 'Invalid email or password'; remains /login; 'Sign In' and fields reenabled; getCurrentSession returns null.
2	Submit sarah@student.utm.my / Wrong123!.	Identical 'Invalid email or password'; no account-existence disclosure; no session.
3	Correct password to admin123 and submit.	'Welcome Back!' then /courses; previous error cleared.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-LOG-02: Generic invalid-credentials errors [Version : 1]
Keywords:	None

Test Case US-111: Missing and malformed login fields cannot authenticate [Version : 1]

Author:	aland

Summary:

Missing and malformed login fields cannot authenticate.

Test design technique: EP (empty email / empty password / malformed email), Error Guessing

Requirements: REQ-A-LOG-03

Preconditions:

Anonymous /login, Network panel open; local authorize test harness available for bypassing native validation.

#:	Step actions:	Expected Results:
1	Submit with Email Address empty/password admin123; then valid Sarah email/empty Password.	Each missing field fails required validity; no signIn request; 'Email Address', 'Password' and 'Sign In' remain displayed.
2	Submit email 'sarah-at-student.utm.my', password admin123.	Native type=email validity fails; no signIn request; exact browser validation wording is browser-dependent.
3	Invoke authorize with email='' and password='admin123', then with valid Sarah email and password='', using local harness.	Each throws 'Missing email or password'; no user is authorized. External login UI maps auth errors to 'Invalid email or password'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-LOG-03: Required login fields [Version : 1]
Keywords:	None

Test Case US-112: Existing sessions bypass login and signup by role [Version : 1]

Author:	aland

Summary:

Existing sessions bypass login and signup by role.

Test design technique: Decision Table (APPROVED/PENDING/ADMIN x /login,/signup)

Requirements: REQ-A-LOG-04, REQ-A-LOG-01

Preconditions:

Three separate browser profiles already signed in as Sarah APPROVED, Alice PENDING and Admin ADMIN with seeded credentials.

#:	Step actions:	Expected Results:
1	In Sarah's profile navigate to /login and /signup separately.	Both end at /courses; 'Courses' visible; no new login/registration form remains.
2	In Alice's profile navigate to /login and /signup separately.	Both end at /waiting-approval with 'Account Pending Approval'.
3	In Admin's profile navigate to /login and /signup separately.	Both finish at /admin/approvals with 'Student Approvals'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-A-LOG-01: Valid credentials route by current role [Version : 1] REQ-A-LOG-04: Session persistence and authenticated-page redirects [Version : 1]
Keywords:	None

Test Case US-113: JWT session survives reload and another tab without exposing password [Version : 1]

Author:	aland

Summary:

JWT session survives reload and another tab without exposing password.

Test design technique: Use Case (session continuation), Error Guessing (password disclosure)

Requirements: REQ-A-LOG-04

Preconditions:

Sarah logged in at /courses in one normal browser profile; developer tools and user.getCurrentSession client available.

#:	Step actions:	Expected Results:
1	Reload /courses, then open /courses in a new tab of the same profile.	Both render authenticated 'Courses'; neither asks for 'Sign In'.
2	Inspect /api/auth/session and user.getCurrentSession.	user.id matches Sarah, role APPROVED, universityId and facultyId match seed; no password/plaintext/hash field is exposed.
3	Inspect authentication session configuration and persisted session cookie existence.	JWT session cookie exists; auth/config.ts specifies strategy 'jwt'. No new database Session row or Remember me control is required by the implemented flow.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-LOG-04: Session persistence and authenticated-page redirects [Version : 1]
Keywords:	None

Test Case US-114: Pending session observes admin approval without signing in again [Version : 1]

Author:	aland	
Summary: Pending session observes admin approval without signing in again. **Test design technique:** State Transition (PENDING -> APPROVED), Use Case (check approval status) **Requirements:** REQ-A-LOG-05, REQ-A-ADM-06		
Preconditions: Alice PENDING signed in at /waiting-approval in profile A; Admin signed in profile B; restore Alice afterward.		
#:	Step actions:	Expected Results:
1	In profile B click Alice's 'Approve' and confirm 'Are you sure you want to approve this user?'.	Alice removed from 'Pending Approvals'; database role APPROVED and approvedAt set.
2	In profile A, without sign out or password entry, navigate to /courses and fetch fresh /api/auth/session.	'Courses' is accessible; fresh session role APPROVED for same Alice id, reflecting current database role.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1] REQ-A-LOG-05: Fresh approval status without a new login [Version : 1]	
Keywords:	None	

1.6.Test Suite : TS-A6 Admin Access and Applicant Review

UC-S2-002 authorization matrix, list filtering/order, student ID inspection, statistics and document-only search/sort. Decision Table, EP, BVA, Use Case.

Test Case US-115: Anonymous and nonadmin users cannot access admin pages or procedures [Version : 1]		
Author:	aland	
Summary: Anonymous and nonadmin users cannot access admin pages or procedures. **Test design technique:** Decision Table (anonymous/APPROVED/PENDING x admin read/approve/reject) **Requirements:** REQ-A-ADM-01		
Preconditions: Original seed; use anonymous, Sarah APPROVED and Alice PENDING profiles. Capture Alice's id, role and approvedAt. Mutations target Alice; run without ADMIN session.		
#:	Step actions:	Expected Results:
1	As anonymous open /admin/approvals; call admin.getPendingUsers, getAllUsers, getDashboardStats, approveUser({userId:AliceId}) and rejectUser({userId:AliceId}).	Page ends at /login showing 'Sign In'; each API request returns UNAUTHORIZED; Alice remains PENDING.
2	As Sarah repeat the page and the five API calls.	Page ends at /courses with 'Courses'; each API returns FORBIDDEN 'Only administrators can perform this action'; no user data or mutation allowed.
3	As Alice repeat the page and the five API calls.	Page ends at /waiting-approval with 'Account Pending Approval'; each API returns FORBIDDEN 'Only administrators can perform this action'; Alice remains PENDING, approvedAt null.
Execution type:	Manual	

Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-01: Admin route and API authorization [Version : 1]
Keywords:	None

Test Case US-116: Pending list excludes approved/admin accounts, orders newest first and handles zero [Version : 1]		
Author:	aland	
Summary:		
Pending list excludes approved/admin accounts, orders newest first and handles zero. **Test design technique:** EP (PENDING/APPROVED/ADMIN), BVA (list size 0/1/2) **Requirements:** REQ-A-ADM-02		
Preconditions:		
Admin signed in; disposable fixture with Alice as sole PENDING, other seeded users unchanged. Can add new PENDING applicant 'Zara QA', zara.qa@example.com, via baseline register with later createdAt; controlled SMTP.		
#:	Step actions:	Expected Results:
1	Open /admin/approvals and inspect admin.getPendingUsers response with Alice sole PENDING.	One Alice card under 'Pending Approvals'; Sarah, John and Admin excluded; response has no password field.
2	Register Zara QA after Alice; refresh admin page and inspect admin.getAllUsers response.	Exactly two pending cards: Zara then Alice (createdAt descending); getAllUsers includes roles but no password field for any user.
3	Approve both disposable pending users as Admin and wait for refreshed queries.	Zero cards; 'All caught up!' and 'There are no pending approvals at the moment.' displayed.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-ADM-02: Pending list and safe user data [Version : 1]	
Keywords:	None	

Test Case US-117: Admin sees applicant identity and opens the stored student ID [Version : 1]		
Author:	aland	
Summary:		
Admin sees applicant identity and opens the stored student ID. **Test design technique:** Use Case (review student identity), EP (pending applicant) **Requirements:** REQ-A-ADM-03		
Preconditions:		
Admin at /admin/approvals; Alice PENDING, original seed URL https://utfs.io/f/example-alice-id.jpg. Seed URL may not host a real image; use a known reachable controlled image URL in Alice's local fixture for rendering if needed and record it.		
#:	Step actions:	Expected Results:
1	Inspect Alice's card.	'Alice Wong', 'alice@student.utm.my', 'University:', 'Universiti Teknologi Malaysia', 'Faculty:', 'Faculty of Computing', 'Registered:', 'Student ID', 'Approve', 'Reject' are displayed; date matches local rendering of createdAt.

2	Click 'View full size →'; inspect new-tab URL.	Opens exact stored studentIdUrl in a new tab; the app does not alter or substitute it. Seed-link availability is a fixture limitation, not evidence of rendering success.
3	Close new tab; click the Student ID image/preview.	Again opens the same stored studentIdUrl; admin page remains at /admin/approvals.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-ADM-03: Review identity details and full-size student ID [Version : 1]	
Keywords:	None	

Test Case US-118: Dashboard role counts refresh after approve and reject [Version : 1]		
Author:	aland	
Summary: Dashboard role counts refresh after approve and reject. **Test design technique:** State Transition (role counts after decisions), Use Case **Requirements:** REQ-A-ADM-04		
Preconditions: Admin signed in; eight original seeded users (1 ADMIN, 6 APPROVED, 1 PENDING) plus newly registered PENDING 'Aland QA', aland.qa.32@example.com. Initial counts: total9, pending2, approved6, admin1; no other mutations.		
#:	Step actions:	Expected Results:
1	Open /admin/approvals; compare four cards to admin.getDashboardStats.	'Pending Approvals'=2, 'Approved Users'=6, 'Total Users'=9, 'Administrators'=1.
2	Approve Alice and confirm; wait for refetch without manual reload.	Counts pending1, approved7, total9, administrators1; Alice card removed.
3	Reject Aland QA, leave reason blank, confirm; wait for refetch.	Counts pending0, approved7, total8, administrators1; 'All caught up!' displayed.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-A-ADM-04: Dashboard totals refresh after decisions [Version : 1]	
Keywords:	None	

Test Case US-119: Documented pending-applicant search and sorting are available [Version : 1]	
Author:	aland
Summary: Documented pending-applicant search and sorting are available. **Test design technique:** Use Case (search/filter alternative), EP (name/email/no match) **Requirements:** REQ-A-ADM-05	
Preconditions: Admin signed in; Alice and newly registered PENDING Zara QA (zara.qa@example.com) exist with distinct dates and universities. Document-only case: current UI has no search/sort controls.	

#:	Step actions:	Expected Results:
1	Open /admin/approvals and locate name/email filtering and date/name/university sorting described in Iteration 2 Use Case 2 step 8.	Document expectation: controls exist. Current page only displays 'Pending Approvals' cards without these controls; record implementation gap if absent; no exact labels are specified by code.
2	If controls exist, filter 'Alice', 'zara.qa@example.com' and 'no-such-applicant', clearing between inputs.	Document expectation: Alice only, Zara only, then zero applicants; reset restores both. Current absence prevents these actions and must not be reported as passed.
3	If available, sort by date, name and university in each supported direction.	Document expectation: stable list ordered by selected field, containing only PENDING applicants. Current code offers only fixed createdAt descending order.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-A-ADM-05: Search/filter and sorting (document-only gap) [Version : 1]	
Keywords:	None	

1.7.Test Suite : TS-A7 Approve Student Registration

UC-S2-002 approval: cancel/confirm, role/timestamp transition, email, invalid targets and sequential replay. State Transition, Decision Table, Error Guessing.

Test Case US-120: Cancel approval leaves applicant pending [Version : 1]		
Author:	aland	
Summary: Cancel approval leaves applicant pending. **Test design technique:** Decision Table (approval confirmation = cancel), State Transition (no transition) **Requirements:** REQ-A-ADM-06		
Preconditions: Admin signed in at /admin/approvals; Alice PENDING, approvedAt null; initial counts and mail capture noted; Network panel open.		
#:	Step actions:	Expected Results:
1	Click Alice's 'Approve'.	Browser confirmation reads 'Are you sure you want to approve this user?'.
2	Choose Cancel and reload.	No admin.approveUser mutation; Alice card still present, role PENDING, approvedAt null; counts and captured mail unchanged.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1]	
Keywords:	None	

Test Case US-121: Confirm approval updates role, timestamp and email [Version : 1]	
Author:	aland
Summary: Confirm approval updates role, timestamp and email. **Test design technique:** State Transition (PENDING -> APPROVED), Use Case main flow	

Requirements: REQ-A-ADM-06, REQ-A-ADM-08

Preconditions:

Admin at /admin/approvals; Alice PENDING, approvedAt null; successful controlled SMTP transport can delay response 2 seconds; record start time and NEXT_PUBLIC_APP_URL=http://localhost:3000.

#:	Step actions:	Expected Results:
1	Click Alice's 'Approve' and confirm 'Are you sure you want to approve this user?'.	While mutation pending, selected button says 'Approving...' and both Approve/Reject buttons disable; no second UI decision can be submitted.
2	Inspect response, database and refreshed list.	success=true, 'Alice Wong has been approved successfully'; role APPROVED, approvedAt between start and completion times; Alice card removed. No success toast is required by current UI.
3	Inspect captured approval email to alice@student.utm.my.	Exact subject starts with U+1F389 (party popper) and a space, followed by 'Your UNIShare Account Has Been Approved!'; body 'Account Approved!' and dashboard link http://localhost:3000/courses.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1] REQ-A-ADM-08: Approval email failure does not undo approval [Version : 1]
Keywords:	None

Test Case US-122: Approve refuses already approved and admin targets [Version : 1]

Author:	aland

Summary:

Approve refuses already approved and admin targets.

Test design technique: Decision Table (target role APPROVED/ADMIN), State Transition invalid

Requirements: REQ-A-ADM-07

Preconditions:

Admin API session; original seed Sarah APPROVED and Admin ADMIN; capture their role/approvedAt and email counts.

#:	Step actions:	Expected Results:
1	Call admin.approveUser({userId:SarahId}).	BAD_REQUEST 'User is not pending approval'; Sarah unchanged; no approval email.
2	Call admin.approveUser({userId:AdminId}).	BAD_REQUEST 'User is not pending approval'; Admin remains ADMIN and approvedAt unchanged; no email.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1]
Keywords:	None

Test Case US-123: Approve rejects nonexistent or empty target ID [Version : 1]

Author:	aland

Summary:

Approve rejects nonexistent or empty target ID.

Test design technique: EP (unknown/empty userId), Error Guessing

Requirements: REQ-A-ADM-07

Preconditions:

Admin API session; ensure userId 'qa-missing-user' does not exist; capture user counts and mail capture.

#:	Step actions:	Expected Results:
1	Call admin.approveUser({userId:'qa-missing-user'}).	NOT_FOUND 'User not found'; no user/count changes and no mail.
2	Call admin.approveUser({userId:''}).	NOT_FOUND 'User not found'; empty string passes z.string but finds no record; no data changes.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1]
Keywords:	None

Test Case US-124: Approval remains effective when approval email fails [Version : 1]

Author:	aland

Summary:

Approval remains effective when approval email fails.

Test design technique: Error Guessing (SMTP failure), State Transition

Requirements: REQ-A-ADM-08, REQ-A-ADM-06, REQ-A-LOG-01

Preconditions:

Admin signed in; Alice PENDING; isolated SMTP stub throws Error('QA SMTP down'); separate anonymous browser ready; restore stub afterward.

#:	Step actions:	Expected Results:
1	Call admin.approveUser({userId:AliceId}) while mail fails.	success=true, 'Alice Wong has been approved successfully'; server logs 'Failed to send approval email:'; no rollback.
2	Inspect Alice then log in as alice@student.utm.my / admin123 in other profile.	Alice APPROVED with approvedAt set; login succeeds and reaches /courses. The document's 'Approved but email failed to send' warning is not implemented and is not claimed as an observed UI message.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1] REQ-A-ADM-08: Approval email failure does not undo approval [Version : 1] REQ-A-LOG-01: Valid credentials route by current role [Version : 1]
Keywords:	None

Test Case US-125: Sequential replay of approval preserves first approval timestamp [Version : 1]

Author:	aland

Summary:

Sequential replay of approval preserves first approval timestamp.

Test design technique: State Transition (PENDING -> APPROVED -> invalid repeat), Error Guessing

Requirements: REQ-A-ADM-07, REQ-A-ADM-06

Preconditions:

Admin API session; Alice initially PENDING; successful SMTP capture; no concurrent calls; record mail count.

#:	Step actions:	Expected Results:
1	Approve Alice once and record approvedAt after success.	'Alice Wong has been approved successfully'; Alice APPROVED; one approval email attempt.
2	After first response completes, approve Alice again.	BAD_REQUEST 'User is not pending approval'; first approvedAt unchanged; no extra approval email; Alice not restored to pending list.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1] REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1]
Keywords:	None

1.8.Test Suite : TS-A8 Reject Student Registration

UC-S2-002 rejection: optional reason, final confirmation, deletion, invalid targets, mail failure and re-registration. EP, Decision Table, State Transition, Use Case, Error Guessing.

Test Case US-126: Reject with reason sends explanation and deletes pending applicant [Version : 1]

Author: aland

Summary:

Reject with reason sends explanation and deletes pending applicant.

Test design technique: State Transition (PENDING -> deleted), Use Case rejection flow

Requirements: REQ-A-ADM-09, REQ-A-ADM-10

Preconditions:

Admin at /admin/approvals; Alice PENDING, captured id and row; SMTP capture succeeds with 2-second delay; no active Alice session.

#:	Step actions:	Expected Results:
1	Click Alice's 'Reject'; enter 'Student ID image is unreadable' at 'Enter rejection reason (optional):'.	Then confirmation 'Are you sure you want to reject this user?' appears; no deletion before final confirmation.
2	Confirm rejection and inspect pending UI/response.	While pending selected action says 'Rejecting...' and both decision buttons disable; success=true, 'Registration for Alice Wong has been rejected'; Alice card removed.
3	Inspect database and captured rejection mail.	No User with Alice's id/email remains; no REJECTED status record; subject 'UNIShare Registration Update', body 'Registration Status: Not Approved', 'Reason:' and 'Student ID image is unreadable'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1] REQ-A-ADM-10: Rejection email is best effort [Version : 1]
Keywords:	None

Test Case US-127: Optional rejection reason accepts blank and canceled prompt [Version : 1]		
Author:	aland	
Summary: Optional rejection reason accepts blank and canceled prompt. **Test design technique:** EP (empty/omitted reason), Decision Table (prompt result x final confirmation) **Requirements:** REQ-A-ADM-09, REQ-A-ADM-10		
Preconditions: Admin page; restore independent Alice PENDING fixture before each row; successful SMTP capture; Network panel open.		
#:	Step actions:	Expected Results:
1	Reject Alice; leave reason empty, choose OK, then confirm final 'Are you sure you want to reject this user?'.	reason='' accepted; 'Registration for Alice Wong has been rejected'; Alice deleted; subject 'UNIShare Registration Update' with no 'Reason:' section.
2	Restore Alice fixture. Reject again; Cancel the optional reason prompt, then confirm the separate final rejection dialog.	Current UI converts null prompt to omitted reason and still proceeds after final confirmation; same success response and deletion; email omits 'Reason:'. Canceling the reason prompt alone is not final cancellation.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1] REQ-A-ADM-10: Rejection email is best effort [Version : 1]	
Keywords:	None	

Test Case US-128: Cancel final rejection confirmation preserves applicant [Version : 1]		
Author:	aland	
Summary: Cancel final rejection confirmation preserves applicant. **Test design technique:** Decision Table (final confirmation = cancel), State Transition (no transition) **Requirements:** REQ-A-ADM-09		
Preconditions: Admin page; Alice PENDING, approvedAt null; user/count snapshots and mail count captured; Network panel open.		
#:	Step actions:	Expected Results:
1	Click 'Reject'; enter 'QA cancellation check' and accept reason prompt.	Final confirmation says 'Are you sure you want to reject this user?'.
2	Choose Cancel on final confirmation, then reload admin page.	No admin.rejectUser request; Alice card remains PENDING with same ID and null approvedAt; counts unchanged; no rejection mail.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1]	
Keywords:	None	

Test Case US-129: Reject refuses approved/admin targets and missing users [Version : 1]	
Author:	aland
Summary:	

Reject refuses approved/admin targets and missing users.

Test design technique: Decision Table (target APPROVED/ADMIN/missing), State Transition invalid

Requirements: REQ-A-ADM-07

Preconditions:

Admin API session; original Sarah APPROVED and Admin ADMIN; userId 'qa-missing-user' absent; capture their rows and mail count.

#:	Step actions:	Expected Results:
1	Call admin.rejectUser({userId:SarahId,reason:'QA'}) then repeat with AdminId.	Each returns BAD_REQUEST 'User is not pending approval'; neither user is deleted; roles and approvedAt unchanged; no email.
2	Call admin.rejectUser({userId:'qa-missing-user'}) then {userId:''} separately.	Each returns NOT_FOUND 'User not found'; no user/count changes and no email.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1]
Keywords:	None

Test Case US-130: Rejection still deletes applicant when mail fails [Version : 1]

Author:	aland

Summary:

Rejection still deletes applicant when mail fails.

Test design technique: Error Guessing (SMTP failure), State Transition (PENDING -> deleted)

Requirements: REQ-A-ADM-10, REQ-A-ADM-09

Preconditions:

Admin API session; Alice PENDING; isolated mail stub throws Error('QA SMTP down'); store Alice's id; restore transport afterward.

#:	Step actions:	Expected Results:
1	Call admin.rejectUser({userId:AliceId,reason:'Unreadable ID'}) with failing SMTP.	Logs 'Failed to send rejection email:' but returns success=true, 'Registration for Alice Wong has been rejected'.
2	Inspect database, refresh pending list, then retry reject for old AliceId.	Alice deleted and absent from 'Pending Approvals'; retry returns NOT_FOUND 'User not found'; no retained REJECTED account.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1] REQ-A-ADM-10: Rejection email is best effort [Version : 1]
Keywords:	None

Test Case US-131: Deleted applicant cannot login and may register again for review [Version : 1]

Author:	aland

Summary:

Deleted applicant cannot login and may register again for review.

Test design technique: State Transition (PENDING -> deleted -> PENDING), Use Case alternative

Requirements: REQ-A-ADM-09, REQ-A-LOG-02, REQ-A-REG-16

Preconditions:

Admin API session; Alice initially PENDING; fresh anonymous browser has never logged in as Alice; controlled SMTP. Record old AliceId. Use new registration password Test1234, not seed admin123.

#:	Step actions:	Expected Results:
1	As Admin reject Alice and confirm; in fresh anonymous profile attempt alice@student.utm.my / admin123 login.	Rejection response 'Registration for Alice Wong has been rejected'; login shows 'Invalid email or password'; no session established.
2	On /signup register name Alice Wong, same alice@student.utm.my email, Test1234/confirmation, seeded UTM/Computing and newly uploaded valid ID.	'Registration Successful!'; exactly one new Alice user with a different id, role PENDING and approvedAt null; email uniqueness does not prevent reapplication after deletion.
3	Log in with alice@student.utm.my / Test1234.	Ends at /waiting-approval with 'Account Pending Approval'; prior rejection does not bypass new admin review.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1] REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1] REQ-A-LOG-02: Generic invalid-credentials errors [Version : 1]
Keywords:	None

2.Test Suite : Karoz - Course, Resource & Timetable Management (Iteration 3 UC1, Iteration 4 UC-S4-005)

Tester: Karoz
Software under test: UniShare (github.com/Seganation/unishare)
Test basis: UniShare_Iteration_3.md Use Case 1 (UC-S3-001) and UniShare_Iteration_4.md UC-S4-005, and the code that implements them: tRPC routers course.ts, resource.ts, timetable.ts; src/components/courses/course-form.tsx, src/components/resources/*, src/components/timetable/*, src/app/(student)/courses/*, src/app/(student)/timetable/page.tsx, src/app/api/uploadthing/core.ts, src/app/api/cron/class-reminders/route.ts, middleware.ts.
Techniques: Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table (DT), State Transition (ST), Use Case / scenario testing (UC), Error Guessing (EG).

Common test data (prisma/seed.ts, all seeded passwords are *admin123*)
Student A - Sarah Ahmed (sarah@student.utm.my), APPROVED, Faculty of Computing. Owns "Data Structures & Algorithms" (John = accepted CONTRIBUTOR) and "Web Development"; accepted VIEWER of "Machine Learning" (owner John); favourites: Data Structures & Algorithms, Web Development, Machine Learning; default timetable "Fall 2024 Schedule" (John = accepted VIEWER, David = PENDING).
Student B - John Doe (john@student.utm.my), APPROVED. Owns "Machine Learning" and "Database Systems"; own timetable "Fall 2024 Schedule" on which Sarah is an accepted CONTRIBUTOR and Lisa has a PENDING invitation.
Student C - Michael Chen (michael@student.utm.my), APPROVED, Faculty of Engineering, no access to Sarah's courses.
Student D - Alice Wong (alice@student.utm.my), role PENDING (not approved).
Other approved students: David Kim, Lisa Martinez, Emily Rodriguez.
Fresh data used in the cases (created by the tester): course "SECJ3343 Software Quality Assurance" owned by Sarah; empty timetable "SQA Week" owned by Sarah; course colours = the 8 swatches of src/lib/course-colors.ts (Purple #a855f7, Blue #3b82f6, Green #10b981, Orange #f97316, Red #ef4444, Pink #ec4899, Teal #14b8a6, Indigo #6366f1).
API checks can be done with the browser DevTools console or a tRPC client logged in as the stated student (procedures course.*, resource.*, timetable.*).

2.1.Test Suite : TS-K1 Create Course (UC-S3-001)

Create-course form and API. Techniques: UC, BVA (title 100, description 500), EP (colour), DT, EG.

Test Case US-132: Create a course with title, description and colour (main flow) **[Version : 1]**		
Author:	karoz	
Summary: Create a course with title, description and colour (main flow). **Test design technique:** Use case (main flow), EP valid classes **Requirements:** REQ-CRS-01, REQ-CRS-02, REQ-CRS-03, REQ-CRS-05		
Preconditions: Student A (sarah@student.utm.my / admin123) logged in, role APPROVED, on /courses.		
#:	Step actions:	Expected Results:
1	Click 'Create New Course'.	Page 'Create New Course' opens at /courses/new with the form and the tip about 4 predefined resource cards.
2	Enter title 'SECJ3343 Software Quality Assurance' and description 'Test design techniques and TestLink'.	Both texts accepted; character counters update.
3	Leave the colour unchanged.	Purple is preselected ('Selected Color: Purple').
4	Click the create button.	Toast 'Course created successfully!'; redirect to /courses; the course is listed under 'My Courses' with a purple header.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-CRS-01: Course title validation [Version : 1] REQ-CRS-02: Course description validation [Version : 1] REQ-CRS-03: Course colour [Version : 1] REQ-CRS-05: Successful creation [Version : 1]
Keywords:	None

Test Case US-133: Title length boundaries (0, 1, 100, 101 characters) [Version : 1]

Author:	karoz

Summary:

Title length boundaries (0, 1, 100, 101 characters).

Test design technique: BVA

Requirements: REQ-CRS-01

Preconditions:

Student A on /courses/new, description empty, colour Purple.

#:	Step actions:	Expected Results:
1	Leave title empty and submit.	Not created; toast 'Please enter a course title'.
2	Enter 1 character 'A' and submit.	Course 'A' created (minimum length 1 accepted) - 'Course created successfully!'.
3	Open /courses/new, enter a 100-character title and submit.	Created; full 100-character title shown in the list.
4	Open /courses/new and type/paste a 101-character title.	Input stops at 100 characters (maxLength); counter shows 100/100.
5	Call course.create via API with a 101-character title.	Rejected by validation (too_big, max 100); no course created.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-CRS-01: Course title validation [Version : 1]
Keywords:	None

Test Case US-134: Whitespace-only and padded title [Version : 1]

Author:	karoz

Summary:

Whitespace-only and padded title.

Test design technique: Error guessing

Requirements: REQ-CRS-01

Preconditions:

Student A on /courses/new.

#:	Step actions:	Expected Results:

1	Enter 5 spaces as title and submit.	Toast 'Please enter a course title'; nothing created (title is trimmed).
2	Enter ' Networks ' (leading/trailing spaces) and submit.	Course created with title 'Networks' (trimmed).
3	Create a course titled 'C++ & Java "quotes"'.	Created; the title is displayed as plain text exactly as typed (no HTML rendering).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-CRS-01: Course title validation [Version : 1]	
Keywords:	None	

Test Case US-135: Description length boundaries (empty, 500, 501 characters) **[Version : 1]**

Author:	karoz	
Summary:		
Description length boundaries (empty, 500, 501 characters). **Test design technique:** BVA **Requirements:** REQ-CRS-02		
Preconditions:		
Student A on /courses/new, valid title entered.		
#:	Step actions:	Expected Results:
1	Leave description empty and submit.	Course created without description.
2	Create another course with a 500-character description.	Created; description saved completely.
3	Type/paste a 501-character description.	Textarea stops at 500 characters (counter 500/500).
4	Call course.create via API with a 501-character description.	Rejected by validation (max 500); nothing created.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-CRS-02: Course description validation [Version : 1]	
Keywords:	None	

Test Case US-136: Colour selection - each swatch **[Version : 1]**

Author:	karoz
Summary:	
Colour selection - each swatch. **Test design technique:** EP (8 swatches, default) **Requirements:** REQ-CRS-03	
Preconditions:	

Student A on /courses/new with title 'Colour test'.

#:	Step actions:	Expected Results:
1	Look at the colour section.	8 swatches: Purple, Blue, Green, Orange, Red, Pink, Teal, Indigo; Purple selected.
2	Click 'Select Teal color', submit.	Course created; course card header/colour is teal (#14b8a6).
3	Repeat for Blue (#3b82f6) and Indigo (#6366f1).	Each course shows the chosen colour; 'Selected Color' label matched the swatch before submit.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-CRS-03: Course colour [Version : 1]
Keywords:	None

Test Case US-137: Colour format validation in the API [Version : 1]

Author:	karoz

Summary:

Colour format validation in the API.

Test design technique: EP / BVA (valid and invalid hex strings)

Requirements: REQ-CRS-03

Preconditions:

Student A logged in; course.create callable from the browser console or a tRPC client.

#:	Step actions:	Expected Results:
1	Call course.create with color '#3B82F6' (upper-case).	Accepted (case-insensitive).
2	Call with color '#3b82f6' (lower-case).	Accepted.
3	Call with '#3B82F' (5 digits) and '#3B82F6A' (7 digits).	Both rejected: 'Invalid color format'.
4	Call with '3B82F6' (no #), '#GGGGGG' and ''.	All rejected: 'Invalid color format'; no course created.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-CRS-03: Course colour [Version : 1]
Keywords:	None

Test Case US-138: Four default resource cards are created [Version : 1]

Author:	karoz

Summary:

Four default resource cards are created.

Test design technique: Decision table (card x file-upload flag), use case

Requirements: REQ-CRS-04, REQ-CRS-05

Preconditions:

Student A just created course 'SECJ3343 Software Quality Assurance'.

#:	Step actions:	Expected Results:
1	Open the course from 'My Courses'.	Course page opens with role badge OWNER and the Resources tab.
2	Count the resource cards.	Exactly 4 cards: Assignments, Tasks, Content, Notes (in this order of creation).
3	Check the file buttons.	'Add Files' is shown on Assignments and Content only; Tasks and Notes have no file upload.
4	Inspect each card.	Cards without files show 'No files uploaded yet' / empty state; no delete button on any default card.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-CRS-04: Four default resource cards [Version : 1] REQ-CRS-05: Successful creation [Version : 1]
Keywords:	None

Test Case US-139: Duplicate course titles are allowed [Version : 1]

Author:	karoz

Summary:

Duplicate course titles are allowed.

Test design technique: Error guessing

Requirements: REQ-CRS-05

Preconditions:

Student A already owns 'SECJ3343 Software Quality Assurance'.

#:	Step actions:	Expected Results:
1	Create a second course with exactly the same title.	Created ('Course created successfully!').
2	Open /courses.	Two separate cards with the same title under 'My Courses', each with its own 4 resource cards.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Low
Requirements	REQ-CRS-05: Successful creation [Version : 1]
Keywords:	None

Test Case US-140: Course creation blocked for visitors and PENDING students [Version : 1]

Author:	karoz

Summary:

Course creation blocked for visitors and PENDING students.

Test design technique: Decision table (login state x role)

Requirements: REQ-CRS-06

Preconditions:		
Browser logged out. Student D (alice@student.utm.my / admin123) has role PENDING.		
#:	Step actions:	Expected Results:
1	Logged out, open /courses/new.	Redirected to /login.
2	Log in as Student D and open /courses/new.	Redirected to /waiting-approval; the form is not shown.
3	Logged in as Student D open /courses and /timetable.	Both redirect to /waiting-approval.
4	Log in as Student A and open /courses/new.	Form is shown (APPROVED allowed).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-CRS-06: Only approved, logged-in students can create courses [Version : 1]	
Keywords:	None	

Test Case US-141: course.create called without a session or with a PENDING session [Version : 1]		
Author:	karoz	
Summary:		
course.create called without a session or with a PENDING session. **Test design technique:** Error guessing (API bypass) **Requirements:** REQ-CRS-06		
Preconditions:		
Student D (PENDING) logged in; separately a logged-out client.		
#:	Step actions:	Expected Results:
1	Logged-out client calls course.create with a valid payload.	Rejected with UNAUTHORIZED; nothing created.
2	As Student D call course.create with a valid payload.	Expected per Iteration 3 A3: 403 FORBIDDEN 'Your account is pending approval'; no course created. (Code only checks the session, so the course is created - defect to report.)
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-CRS-06: Only approved, logged-in students can create courses [Version : 1]	
Keywords:	None	

2.2.Test Suite : TS-K2 Manage Courses

List, open, edit, delete and favourite a course. Techniques: DT (role x action), EP, ST, EG.

Test Case US-142: Course list shows owned and shared courses only [Version : 1]	
Author:	karoz
Summary:	
Course list shows owned and shared courses only.	

Test design technique: EP (owned / accepted shared / pending / other)

Requirements: REQ-MAN-01

Preconditions:

Seed data. Student A owns Data Structures & Algorithms and Web Development and is accepted VIEWER of Machine Learning (owner John).

#:	Step actions:	Expected Results:
1	Log in as Student A and open /courses.	'My Courses' contains Data Structures & Algorithms and Web Development.
2	Look at 'Shared With Me'.	Contains Machine Learning (owned by John). Database Systems / Operating Systems are not listed.
3	Log in as an approved student E who has no courses and open /courses.	Message 'No courses yet' with the create-course call to action.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-MAN-01: Course list: owned and shared [Version : 1]
Keywords:	None

Test Case US-143: Course access by role [Version : 1]

Author:	karoz

Summary:

Course access by role.

Test design technique: Decision table (owner / contributor / viewer / outsider / unknown id)

Requirements: REQ-MAN-02

Preconditions:

Course 'Data Structures & Algorithms' (owner Sarah, John accepted CONTRIBUTOR). Michael has no relation to it.

#:	Step actions:	Expected Results:
1	Sarah opens the course.	Opens; badge OWNER; 'Add Custom Resource' and the three-dots menu (Course Settings, Manage Collaborators) visible.
2	John opens the course.	Opens; badge CONTRIBUTOR; 'Add Custom Resource' visible; no owner menu.
3	Sarah opens Machine Learning (she is VIEWER).	Opens; badge VIEWER; no 'Add Custom Resource' button.
4	Michael opens the Data Structures course URL.	Access refused: 'You don't have access to this course' (no course content shown).
5	Open /courses/does-not-exist as Sarah.	'Course Not Found' page with a 'Back to Courses' button (API: 'Course not found').

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-MAN-02: Course access control [Version : 1]

Keywords:	None

Test Case US-144: Owner edits title, description and colour [Version : 1]		
Author:	karoz	
Summary: Owner edits title, description and colour. **Test design technique:** Use case, EP **Requirements:** REQ-MAN-03		
Preconditions: Student A owns 'SECJ3343 Software Quality Assurance' (Purple).		
#:	Step actions:	Expected Results:
1	Course page > three-dots > Course Settings.	Settings page with tabs General and Danger Zone.
2	Change title to 'SECJ3343 SQA', description to 'Updated', colour to Green; save.	Toast 'Course updated successfully!'.
3	Return to /courses.	Card shows the new title and the green colour; resource cards unchanged (still 4).
4	Set title to 101 characters via API (course.update).	Rejected (max 100); course keeps its title.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-MAN-03: Edit course (owner only) [Version : 1]	
Keywords:	None	

Test Case US-145: Non-owner cannot edit or open settings [Version : 1]		
Author:	karoz	
Summary: Non-owner cannot edit or open settings. **Test design technique:** Decision table (role x action), negative **Requirements:** REQ-MAN-03		
Preconditions: John is CONTRIBUTOR and Sarah is VIEWER of other courses (seed); Michael has no access.		
#:	Step actions:	Expected Results:
1	John opens /courses/<DSA id>/settings.	Page shows 'Access Denied' and 'Only the course owner can access settings.'.
2	John calls course.update on the DSA course.	FORBIDDEN 'Only the course owner can update course details'; data unchanged.
3	Sarah (VIEWER of Machine Learning) calls course.update on it.	FORBIDDEN with the same message.
4	Call course.update with an unknown id.	NOT_FOUND 'Course not found'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-MAN-03: Edit course (owner only) [Version : 1]
Keywords:	None

Test Case US-146: Clearing the description when editing a course [Version : 1]		
Author:	karoz	
Summary: Clearing the description when editing a course. **Test design technique:** Error guessing **Requirements:** REQ-MAN-03		
Preconditions: Student A's course 'SECJ3343 SQA' has description 'Updated'.		
#:	Step actions:	Expected Results:
1	Open Course Settings > General, delete the whole description text and save.	'Course updated successfully!'.
2	Reload the course/list.	Description is empty/removed. (Code sends undefined for an empty description, so the old text may remain - defect to report.)
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-MAN-03: Edit course (owner only) [Version : 1]	
Keywords:	None	

Test Case US-147: Delete course - confirm and cancel [Version : 1]		
Author:	karoz	
Summary: Delete course - confirm and cancel. **Test design technique:** State transition (exists -> deleted), use case **Requirements:** REQ-MAN-04		
Preconditions: Student A owns course 'To delete' with 1 custom resource; John is an accepted collaborator; Student A favourited it.		
#:	Step actions:	Expected Results:
1	Course Settings > Danger Zone > 'Delete Course Permanently'.	Browser confirm: 'Are you sure you want to delete "To delete"? This action cannot be undone and will delete all resources and collaborator access.'.
2	Click Cancel.	Course still exists; nothing changed.
3	Repeat and click OK.	Toast 'Course deleted successfully'; redirected away; course no longer in 'My Courses'.

4	Log in as John.	Course no longer under 'Shared With Me'; opening its old URL shows 'Course Not Found'.
5	Open Sarah's favourites / timetable course dropdown.	Deleted course not listed (favourite removed with the course).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-MAN-04: Delete course (owner only) [Version : 1]	
Keywords:	None	

Test Case US-148: Non-owner cannot delete a course [Version : 1]		
Author:	karoz	
Summary: Non-owner cannot delete a course. **Test design technique:** Decision table, negative **Requirements:** REQ-MAN-04		
Preconditions: John is CONTRIBUTOR of Sarah's 'Data Structures & Algorithms'.		
#:	Step actions:	Expected Results:
1	John calls course.delete with the course id.	FORBIDDEN 'Only the course owner can delete the course'; course still exists.
2	John opens the course page.	No settings/delete entry point is offered.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-MAN-04: Delete course (owner only) [Version : 1]	
Keywords:	None	

Test Case US-149: Toggle favourite on and off [Version : 1]		
Author:	karoz	
Summary: Toggle favourite on and off. **Test design technique:** State transition (not favourite <-> favourite) **Requirements:** REQ-MAN-05		
Preconditions: Student A has not favourited the new SQA course.		
#:	Step actions:	Expected Results:
1	Click the star on the course page.	Toast 'Added to favorites'; star filled yellow.
2	Open Timetable > Add Class.	The course now appears in the course dropdown.

3	Click the star again.	Toast 'Removed from favorites'; star empty; course no longer in the Add Class dropdown.
4	Student B opens the same course (if shared) and checks the star.	Student B's star is independent of Student A's.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-MAN-05: Favourite toggle [Version : 1]	
Keywords:	None	

2.3.Test Suite : TS-K3 Resources & Files

Custom resource cards, permissions, deadlines and file uploads. Techniques: BVA, DT, EP, EG.

Test Case US-150: Owner creates a custom resource card [Version : 1]		
Author:	karoz	
Summary: Owner creates a custom resource card. **Test design technique:** Use case (main flow) **Requirements:** REQ-RES-01, REQ-RES-02		
Preconditions: Student A owns SECJ3343 SQA course (4 default cards).		
#:	Step actions:	Expected Results:
1	Click 'Add Custom Resource'.	Dialog 'Create Custom Resource' opens; 'Allow File Uploads' is ON, 'Set a Deadline' OFF.
2	Enter title 'Midterm Exam Study Guide' and description 'Chapters 1-5'; click create.	Toast 'Resource created successfully!'; dialog closes.
3	Look at the grid.	5 cards; the new card 'Midterm Exam Study Guide' with a delete button and 'Add Files' button.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-RES-01: Create custom resource card [Version : 1] REQ-RES-02: Who may create or update resources [Version : 1]	
Keywords:	None	

Test Case US-151: Resource title and description boundaries [Version : 1]	
Author:	karoz
Summary: Resource title and description boundaries. **Test design technique:** BVA (title 0/1/100/101; description 500/501) **Requirements:** REQ-RES-01	

Preconditions: Student A in the 'Create Custom Resource' dialog.		
#:	Step actions:	Expected Results:
1	Leave title empty and submit.	Not created (required field / toast 'Please enter a title').
2	Create resource with a 1-character title and one with a 100-character title.	Both created.
3	Type a 101-character title.	Field stops at 100 characters.
4	Call resource.create with a 101-character title and with a 501-character description.	Both rejected by validation; nothing created.
5	Create a resource with a 500-character description.	Created.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-RES-01: Create custom resource card [Version : 1]	
Keywords:	None	

Test Case US-152: Deadline states: future, within 3 days, overdue [Version : 1]		
Author:	karoz	
Summary: Deadline states: future, within 3 days, overdue. **Test design technique:** BVA/EP (deadline = +4 days, +2 days, -1 day) **Requirements:** REQ-RES-07, REQ-RES-01		
Preconditions: Student A owns the SQA course; today's date known.		
#:	Step actions:	Expected Results:
1	Create resource 'D1' with 'Set a Deadline' = today + 4 days.	Card shows 'Due: <date> at <time>' in neutral colour, no 'Soon' badge.
2	Create resource 'D2' with deadline today + 2 days.	Card shows 'Due: ...' with amber style and badge 'Soon'.
3	Update 'D3' via resource.update with a deadline yesterday.	Card shows 'Overdue: <date>' in red, no 'Soon' badge.
4	Create a resource without a deadline.	No deadline badge.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-RES-01: Create custom resource card [Version : 1] REQ-RES-07: Deadline display [Version : 1]	
Keywords:	None	

Test Case US-153: Create/update permission matrix [Version : 1]		
Author:	karoz	
Summary: Create/update permission matrix. **Test design technique:** Decision table (role x action) **Requirements:** REQ-RES-02		
Preconditions: Course DSA: owner Sarah, John accepted CONTRIBUTOR. Course Machine Learning: owner John, Sarah accepted VIEWER. Michael unrelated.		
#:	Step actions:	Expected Results:
1	John (CONTRIBUTOR) creates a custom resource in DSA.	Allowed - 'Resource created successfully!'.
2	John updates the title of a resource in DSA.	Allowed.
3	Sarah (VIEWER) opens Machine Learning.	No 'Add Custom Resource' button.
4	Sarah calls resource.create for Machine Learning.	FORBIDDEN 'Only course owner and contributors can create resources'.
5	Sarah calls resource.update on a Machine Learning resource.	FORBIDDEN 'Only course owner and contributors can update resources'.
6	Michael calls resource.create for DSA; call resource.create with an unknown courseId.	FORBIDDEN (same message); 'Course not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-RES-02: Who may create or update resources [Version : 1]	
Keywords:	None	

Test Case US-154: Delete resource - owner only, custom cards only in the UI [Version : 1]		
Author:	karoz	
Summary: Delete resource - owner only, custom cards only in the UI. **Test design technique:** Decision table (role x card type) **Requirements:** REQ-RES-03		
Preconditions: Course DSA: owner Sarah, John CONTRIBUTOR; it has a CUSTOM card 'Extra' and the 4 default cards.		
#:	Step actions:	Expected Results:
1	Sarah looks at the default cards (Assignments, Tasks, Content, Notes).	No delete button on them.
2	Sarah clicks delete on 'Extra'.	Confirm 'Are you sure you want to delete this resource?'; Cancel keeps the card.
3	Click delete again and confirm.	Toast 'Resource deleted successfully'; card removed.

4	John (CONTRIBUTOR) opens the course.	No delete button on any card.
5	John calls resource.delete for a resource of DSA.	FORBIDDEN 'Only the course owner can delete resources'; resource still exists.
6	Call resource.delete with an unknown id.	NOT_FOUND 'Resource not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-RES-03: Delete resource (owner only) [Version : 1]	
Keywords:	None	

Test Case US-155: Attach files to a card that allows uploads [Version : 1]		
Author:	karoz	
Summary:		
Attach files to a card that allows uploads. **Test design technique:** Use case, DT (allowFiles x role) **Requirements:** REQ-RES-04		
Preconditions:		
Student A owns the SQA course; Assignments card allows files. File week1.pdf (1 MB) available. John is CONTRIBUTOR of DSA.		
#:	Step actions:	Expected Results:
1	Click 'Add Files' on Assignments, drop week1.pdf and upload.	Progress shown; toast 'Successfully uploaded 1 file!' then 'Files added successfully'.
2	Look at the card.	week1.pdf listed as a file link; the button now reads 'Add More Files'.
3	John uploads a file to a DSA card that allows files.	Allowed; file listed.
4	A VIEWER (Sarah in Machine Learning) looks at a card.	No 'Add Files' button; existing files can be opened/downloaded.
5	Sarah calls resource.addFile for a Machine Learning resource.	FORBIDDEN 'You don't have permission to add files to this resource'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-RES-04: Attach files to a card [Version : 1]	
Keywords:	None	

Test Case US-156: File upload refused for cards without uploads and invalid URLs [Version : 1]	
Author:	karoz
Summary:	
File upload refused for cards without uploads and invalid URLs. **Test design technique:** EP (allowFiles false / invalid URL), error guessing	

Requirements: REQ-RES-04

Preconditions:

Student A owns the SQA course; Tasks card has allowFiles = false; Assignments allows files.

#:	Step actions:	Expected Results:
1	Open the Tasks card.	No 'Add Files' button is offered.
2	Call resource.addFile on the Tasks card.	BAD_REQUEST 'This resource does not allow file uploads'.
3	Call resource.addFile on Assignments with fileUrls 'not-a-url'.	Rejected by validation (invalid url); no file stored.
4	Call resource.addFile with an array of two valid URLs.	Both URLs added; card lists 2 files.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-RES-04: Attach files to a card [Version : 1]
Keywords:	None

Test Case US-157: Upload limits: size, type and count [Version : 1]

Author:	karoz

Summary:

Upload limits: size, type and count.

Test design technique: BVA (16MB-1, 16MB, 16MB+1; 10/11 files; video 64MB; zip 32MB) and EP (types)

Requirements: REQ-RES-05

Preconditions:

Student A on 'Add Files' of an upload-enabled card. Test files prepared: pdf 15.9MB, pdf 16MB, pdf 16.1MB, mp4 64MB, mp4 65MB, zip 32MB, zip 33MB, .exe 1MB, 10 and 11 small PDFs.

#:	Step actions:	Expected Results:
1	Upload the 15.9MB and the 16MB PDF.	Both accepted.
2	Upload the 16.1MB PDF.	Rejected: toast 'Failed to upload files: ...' (file too large); file marked error.
3	Upload the 64MB mp4, then the 65MB mp4.	64MB accepted; 65MB rejected.
4	Upload the 32MB zip, then the 33MB zip.	32MB accepted; 33MB rejected.
5	Upload the .exe file.	Rejected (type not allowed).
6	Upload 10 small PDFs together, then 11.	10 accepted; 11 rejected (max file count).
7	Log out and call the upload endpoint courseResourceUploader.	Rejected 'Unauthorized'.

Execution type:	Manual
Estimated exec. duration (min):	

Importance:	High
Requirements	REQ-RES-05: Upload size, type and count limits [Version : 1]
Keywords:	None

Test Case US-158: Remove a file from a card [Version : 1]		
Author:	karoz	
Summary: Remove a file from a card. **Test design technique:** DT (role x action) **Requirements:** REQ-RES-06		
Preconditions: Assignments card of the SQA course holds files A and B; John is not a collaborator; Sarah (course DSA) is not CONTRIBUTOR of SQA.		
#:	Step actions:	Expected Results:
1	Owner calls resource.removeFile with the URL of file A.	Card now lists only file B; A is gone.
2	A VIEWER or outsider calls resource.removeFile for file B.	FORBIDDEN 'You don't have permission to remove files from this resource'; B stays.
3	Owner calls removeFile with a valid URL that is not on the card.	No change, no error; list unchanged.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-RES-06: Remove a file [Version : 1]	
Keywords:	None	

2.4.Test Suite : TS-K4 Timetable (UC-S4-005)

Timetables and weekly classes. Techniques: UC, BVA (time ranges, 3 per day, names), DT (roles), ST (invitation).

Test Case US-159: Create a timetable (first-time and name boundaries) [Version : 1]		
Author:	karoz	
Summary: Create a timetable (first-time and name boundaries). **Test design technique:** Use case + BVA (name 0/1/100/101, description 500/501) **Requirements:** REQ-TT-01		
Preconditions: Approved student E without any timetable (or Student A for the boundary steps).		
#:	Step actions:	Expected Results:
1	Student E opens /timetable.	'No Timetables Yet' and button 'Create Your First Timetable'.
2	Create timetable 'SQA Week' with description 'Semester 1'.	Timetable created and listed under 'My Timetables'; calendar week view shown (Monday-Sunday columns).

3	As Student A open Create Timetable and submit an empty name.	Not submitted (required field); API 'Name is required'.
4	Create timetables named with 1 and with 100 characters.	Both created.
5	Type 101 characters in the name; call timetable.create with a 501-character description.	Name field stops at 100; API rejects the 501-character description.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-01: Create timetable [Version : 1]	
Keywords:	None	

Test Case US-160: Set, keep and lose the default timetable [Version : 1]		
Author:	karoz	
Summary: Set, keep and lose the default timetable. **Test design technique:** State transition (not default -> default -> cleared) **Requirements:** REQ-TT-02		
Preconditions: Student A has default 'Fall 2024 Schedule' and the new owned timetable 'SQA Week'.		
#:	Step actions:	Expected Results:
1	Open /timetable.	'Fall 2024 Schedule' is selected first and has the 'Default' badge.
2	Select 'SQA Week' > Timetable Settings > 'Set as Default'.	Button becomes 'Currently Default' (disabled); badge 'Default' moves to 'SQA Week'.
3	Reload /timetable.	'SQA Week' opens first.
4	Delete 'SQA Week' (Delete Timetable, confirm).	Default is cleared; /timetable falls back to the first owned timetable.
5	Student C calls timetable.setDefaultTimetable with Sarah's timetable id.	FORBIDDEN 'You don't have access to this timetable'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-TT-02: Default timetable [Version : 1]	
Keywords:	None	

Test Case US-161: Add a weekly class to the timetable (main flow) [Version : 1]	
Author:	karoz
Summary: Add a weekly class to the timetable (main flow). **Test design technique:** Use case, EP valid classes	

Requirements: REQ-TT-03, REQ-TT-04

Preconditions:

Student A owns empty timetable 'SQA Week' (selected); favourites include Data Structures & Algorithms.

#:	Step actions:	Expected Results:
1	Click 'Add Class'.	Modal 'Add Class' with Course, Title, Day of Week, Start Time, End Time, Location (Optional).
2	Open the Course dropdown.	Only favourites listed (Data Structures & Algorithms, Web Development, Machine Learning); hint 'Only your favorited courses are shown'.
3	Select Data Structures & Algorithms, title 'Application Development Lecture', Monday, 08:00-10:00, location 'Room 301'; click 'Add Class'.	Modal closes without error.
4	Look at the week view.	Class shown on Monday 08:00-10:00 in the course colour (blue); clicking it opens 'Class Details' with Course, Day Monday, Time 08:00 - 10:00, Location Room 301.
5	Move to the next week.	The same class appears again on Monday (weekly recurring).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-TT-03: Only favourite courses can be scheduled [Version : 1] REQ-TT-04: Class event data [Version : 1]
Keywords:	None

Test Case US-162: Add Class with no favourite courses [Version : 1]

Author:	karoz

Summary:

Add Class with no favourite courses.

Test design technique: EP (no favourites)

Requirements: REQ-TT-03

Preconditions:

Student E owns a timetable but has no favourite courses.

#:	Step actions:	Expected Results:
1	Click 'Add Class'.	Notice 'No Favorite Courses' / 'You need to favorite courses before adding them to your timetable'; no form.
2	Click 'Go to Courses'.	Modal closes.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-TT-03: Only favourite courses can be scheduled [Version : 1]
Keywords:	None

Test Case US-163: API refuses classes for non-favourite courses [Version : 1]		
Author:	karoz	
Summary: API refuses classes for non-favourite courses. **Test design technique:** Error guessing (API bypass) **Requirements:** REQ-TT-03		
Preconditions: Student A owns 'SQA Week'; course 'SECJ3343 SQA' is NOT a favourite of A.		
#:	Step actions:	Expected Results:
1	Call timetable.createEvent for 'SQA Week' with that course, Monday 13:00-14:00.	BAD_REQUEST 'You can only add favorited courses to your timetable'; no event.
2	Favourite the course (star) and repeat the call.	Event created.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-TT-03: Only favourite courses can be scheduled [Version : 1]	
Keywords:	None	

Test Case US-164: Class title and location length boundaries [Version : 1]		
Author:	karoz	
Summary: Class title and location length boundaries. **Test design technique:** BVA (title 0/1/100/101, location 100/101) **Requirements:** REQ-TT-04		
Preconditions: Student A, 'SQA Week' selected, favourite course available.		
#:	Step actions:	Expected Results:
1	Leave Title empty.	'Add Class' stays disabled (title required).
2	Add classes with a 1-character title and with a 100-character title on different days.	Both added.
3	Type 101 characters into Title and Location.	Both inputs stop at 100 characters.
4	Call timetable.createEvent with a 101-character title, then with a 101-character location.	Both rejected by validation.
5	Call timetable.createEvent with dayOfWeek -1 and 7.	Both rejected (valid range 0-6); dayOfWeek 0 (Sunday) and 6 (Saturday) accepted.
6	Call timetable.createEvent with startTime '9:00' and '0900'.	Rejected 'Invalid time format (HH:MM)'.
Execution type:	Manual	

Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-TT-04: Class event data [Version : 1]
Keywords:	None

Test Case US-165: End time must be after start time [Version : 1]		
Author:	karoz	
Summary:		
End time must be after start time. **Test design technique:** BVA (end = start-1min, = start, = start+1min) and error guessing **Requirements:** REQ-TT-05		
Preconditions:		
Student A in 'Add Class' with course and title filled, Tuesday, start 10:00.		
#:	Step actions:	Expected Results:
1	Set end time 09:59.	Red message 'End time must be after start time'; 'Add Class' disabled.
2	Set end time 10:00.	Same message and disabled (equal times invalid).
3	Set end time 10:01.	No message; 'Add Class' enabled; class (10:00-10:01) can be added.
4	Call timetable.createEvent with startTime '10:00', endTime '09:00'.	Expected: rejected. (API has no check and stores it - defect to report.)
5	Call timetable.createEvent with startTime '99:99'.	Expected: rejected as invalid time. (Only the regexp ^\d{2}:\d{2}$ is checked - defect to report.)
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-05: End time must be after start time [Version : 1]	
Keywords:	None	

Test Case US-166: Time conflict detection boundaries [Version : 1]		
Author:	karoz	
Summary:		
Time conflict detection boundaries. **Test design technique:** BVA (overlap by 1 minute, back-to-back, other day) **Requirements:** REQ-TT-06		
Preconditions:		
'SQA Week' has 'App Dev Lecture' Monday 08:00-10:00 (course DSA).		
#:	Step actions:	Expected Results:
1	Add class Monday 09:00-11:00 (Web Development).	Message 'Time conflict detected' and 'Overlaps with App Dev Lecture (08:00 - 10:00)'; 'Add Class' disabled.

2	Change to Monday 09:59-11:00.	Still a conflict (1-minute overlap).
3	Change to Monday 10:00-12:00.	No conflict (back-to-back allowed); class can be added.
4	Change to Monday 06:00-08:00.	No conflict (ends exactly at 08:00).
5	Change the day to Tuesday 08:00-10:00.	No conflict on another weekday; can be added.
6	Add a class fully inside 08:30-09:00 on Monday.	Conflict reported (contained interval).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-06: Time conflict detection [Version : 1]	
Keywords:	None	

Test Case US-167: Maximum 3 classes per course per day [Version : 1]		
Author:	karoz	
Summary: Maximum 3 classes per course per day. **Test design technique:** BVA (0, 1, 2, 3 existing classes) **Requirements:** REQ-TT-07, REQ-TT-06		
Preconditions: 'SQA Week' empty; favourite course Web Development.		
#:	Step actions:	Expected Results:
1	Add Web Development class Wednesday 08:00-09:00.	Added (0 existing).
2	Open Add Class again, pick Web Development, Wednesday.	Info 'This course has 1 class on this day. You can add 2 more.'.
3	Add Wednesday 09:00-10:00; then open the form again.	Info 'This course has 2 classes on this day. You can add 1 more.'.
4	Add Wednesday 10:00-11:00 (the 3rd).	Added.
5	Open the form for Web Development, Wednesday, 12:00-13:00.	Orange message 'Maximum 3 classes per course per day. This course already has 3 classes on this day.'; 'Add Class' disabled.
6	Select Thursday for the same course.	Message disappears; class can be added.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-06: Time conflict detection [Version : 1] REQ-TT-07: Maximum 3 classes per course per day [Version : 1]	
Keywords:	None	

Test Case US-168: Edit a class in the details dialog [Version : 1]		
Author:	karoz	
Summary: Edit a class in the details dialog. **Test design technique:** Use case (alternative flow A1), BVA **Requirements:** REQ-TT-08, REQ-TT-05, REQ-TT-06		
Preconditions: 'SQA Week' has 'App Dev Lecture' Monday 08:00-10:00 and 'Lab' Monday 10:00-12:00.		
#:	Step actions:	Expected Results:
1	Click 'App Dev Lecture' > Edit Class.	Dialog 'Edit Class' with current values.
2	Change location to 'Room 302' and Save Changes.	Saved; Class Details shows Room 302.
3	Change end time to 10:01 (overlaps 'Lab').	'Time conflict detected' / 'Overlaps with Lab (10:00 - 12:00)'; Save Changes disabled.
4	Change end time to 10:00 again.	No conflict (the class itself is ignored; back-to-back OK); Save works.
5	Change end time before start time.	'End time must be after start time'; Save disabled.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-TT-05: End time must be after start time [Version : 1] REQ-TT-06: Time conflict detection [Version : 1] REQ-TT-08: Edit and move a class [Version : 1]	
Keywords:	None	

Test Case US-169: Drag and resize a class in the calendar [Version : 1]		
Author:	karoz	
Summary: Drag and resize a class in the calendar. **Test design technique:** Error guessing (UI gestures) **Requirements:** REQ-TT-08		
Preconditions: 'SQA Week' has Monday 08:00-10:00 'App Dev Lecture' and Tuesday 08:00-10:00 'Lab'.		
#:	Step actions:	Expected Results:
1	Drag 'App Dev Lecture' to Wednesday 08:00.	Moved; class now on Wednesday 08:00-10:00.
2	Drag 'Lab' to overlap the lecture on Wednesday.	Alert 'Cannot move event: Time conflict with another class'; Lab stays on Tuesday.
3	Resize the lecture so it overlaps another class on its day.	Alert 'Cannot resize event: Time conflict with another class'; unchanged.

4	Resize it to 08:00-11:00 where free.	Saved; Class Details shows 08:00 - 11:00.
5	As a VIEWER try to drag a class.	Not draggable / no change.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-TT-08: Edit and move a class [Version : 1]	
Keywords:	None	

Test Case US-170: Delete a class [Version : 1]		
Author:	karoz	
Summary: Delete a class. **Test design technique:** Use case (alternative flow A2), state transition **Requirements:** REQ-TT-09		
Preconditions: 'SQA Week' has class 'Lab' on Tuesday.		
#:	Step actions:	Expected Results:
1	Click 'Lab' > Delete.	Confirmation 'Delete this class?' with 'Yes, Delete' and 'Cancel'.
2	Click Cancel.	Class still on the calendar.
3	Delete again and click 'Yes, Delete'.	Class disappears from this and all other weeks.
4	Call timetable.deleteEvent with the deleted id.	NOT_FOUND 'Event not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-TT-09: Delete a class [Version : 1]	
Keywords:	None	

Test Case US-171: Timetable role matrix (owner, contributor, viewer, outsider) [Version : 1]		
Author:	karoz	
Summary: Timetable role matrix (owner, contributor, viewer, outsider). **Test design technique:** Decision table (role x action) **Requirements:** REQ-TT-10, REQ-TT-08, REQ-TT-09		
Preconditions: Seed: John's 'Fall 2024 Schedule' - Sarah CONTRIBUTOR (accepted). Sarah's 'Fall 2024 Schedule' - John VIEWER (accepted). Michael has no access to either.		
#:	Step actions:	Expected Results:

1	Sarah opens John's timetable under 'Shared with Me'.	Role shown as Contributor; 'Add Class' available; she can add, edit and delete a class.
2	John opens Sarah's timetable under 'Shared with Me'.	Role shown as Viewer; no 'Add Class'; classes only viewable.
3	John calls timetable.createEvent / updateEvent / deleteEvent on Sarah's timetable.	FORBIDDEN 'You don't have permission to add events to this timetable' / '... update this event' / '... delete this event'.
4	Michael calls timetable.getById for Sarah's timetable.	FORBIDDEN 'You don't have access to this timetable'.
5	Call timetable.getById with an unknown id.	NOT_FOUND 'Timetable not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-08: Edit and move a class [Version : 1] REQ-TT-09: Delete a class [Version : 1] REQ-TT-10: Timetable permissions [Version : 1]	
Keywords:	None	

Test Case US-172: Invite a student to a timetable (search length, role, duplicates) [Version : 1]		
Author:	karoz	
Summary: Invite a student to a timetable (search length, role, duplicates). **Test design technique:** BVA (query 2/3 chars), EP (roles), error guessing **Requirements:** REQ-TT-11		
Preconditions: Student A owns 'SQA Week'. Alice (PENDING) and Emily are in the database; David already has a PENDING invitation to Sarah's 'Fall 2024 Schedule'.		
#:	Step actions:	Expected Results:
1	Select 'SQA Week' > Share.	Dialog with 'Invite as' Viewer/Contributor and 'Search for students' (min 3 characters).
2	Type 'em'.	No search is run; hint 'Type at least 3 characters to search'.
3	Type 'ema'.	Results include Emily Rodriguez; Student A herself is not listed.
4	Search 'ali'.	Alice Wong (PENDING, not approved) is not listed.
5	Choose Contributor and click Invite for Emily.	Invitation created (PENDING); Emily gets notification 'New timetable invitation: Sarah Ahmed invited you to collaborate on SQA Week'.
6	Call timetable.inviteCollaborator again for Emily.	BAD_REQUEST 'User already invited'.
7	Call inviteCollaborator with Sarah's own user id.	BAD_REQUEST 'You can't invite yourself'.

8	John (not owner) calls inviteCollaborator for 'SQA Week'.	FORBIDDEN 'Only the owner can invite collaborators'.
9	Call timetable.searchUsers with query 'ab'.	Rejected 'Search query must be at least 3 characters'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-11: Invite students to a timetable [Version : 1]	
Keywords:	None	

Test Case US-173: Timetable invitation lifecycle: accept, reject, leave [Version : 1]

Author:	karoz	
Summary:		
Timetable invitation lifecycle: accept, reject, leave. **Test design technique:** State transition (PENDING -> ACCEPTED/REJECTED; ACCEPTED -> left) **Requirements:** REQ-TT-12, REQ-TT-02		
Preconditions:		
Sarah's timetable has a PENDING invitation for David (seed); Lisa has a PENDING invitation on John's timetable; Emily was invited to 'SQA Week' as Contributor.		
#:	Step actions:	Expected Results:
1	David opens /courses.	'Timetable Invitations' lists Sarah's 'Fall 2024 Schedule' with Accept and Reject.
2	David clicks Accept.	Timetable appears under 'Shared with Me' in /timetable; Sarah gets 'Invitation accepted: David Kim accepted your invitation to Fall 2024 Schedule'.
3	Lisa clicks Reject.	Invitation disappears; John gets 'Invitation declined'; timetable not shared with Lisa.
4	David sets the shared timetable as default and then Timetable Settings > Leave Timetable > Confirm Leave.	Timetable removed from 'Shared with Me' and the default is cleared.
5	Sarah (owner) calls timetable.leaveTimetable on her own timetable.	BAD_REQUEST 'You cannot leave a timetable you own. Delete it instead.'.
6	David calls acceptInvitation again for the already handled invitation.	Fails (no PENDING invitation); status is not changed.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-02: Default timetable [Version : 1] REQ-TT-12: Accept, reject and leave a shared timetable [Version : 1]	
Keywords:	None	

Test Case US-174: Class reminder is created at the configured time [Version : 1]

Author:	karoz

Summary:

Class reminder is created at the configured time.

Test design technique: EP / BVA (minutes until class = reminder time -2, -1, 0, +1, +2), use case

Requirements: REQ-TT-13

Preconditions:

Student A has a class today starting at HH:MM = now + 10 min on 'SQA Week' (default timetable), reminder timing at its default of 10 minutes. CRON_SECRET known.

#:	Step actions:	Expected Results:
1	GET /api/cron/class-reminders without Authorization header.	401 'Unauthorized'.
2	GET it with 'Authorization: Bearer <CRON_SECRET>'.	JSON success; message 'Sent 1 reminder(s)' (or more).
3	Student A opens the notification bell.	Notification 'Class starting in 10 minutes' with message '<course> at <HH:MM>'.
4	Call the endpoint again in the same minute.	No duplicate notification is created for the same class that day.
5	Change the class to start in 13 minutes (3 min off) and call the endpoint.	No reminder (outside +-1 minute of the 10-minute setting).
6	Set reminder timing to 5 minutes (Settings) with the class 5 minutes away and call again.	Reminder 'Class starting in 5 minutes' created.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-TT-13: Class reminders [Version : 1]
Keywords:	None

Test Case US-175: Rename and delete a timetable (owner only) **[Version : 1]**

Author:	karoz

Summary:

Rename and delete a timetable (owner only).

Test design technique: Decision table (role x action), state transition

Requirements: REQ-TT-14, REQ-TT-02

Preconditions:

Student A owns 'SQA Week' with 2 classes and Emily as accepted Contributor; 'SQA Week' is A's default and also Emily's default.

#:	Step actions:	Expected Results:
1	Emily calls timetable.update (new name) on 'SQA Week'.	FORBIDDEN 'Only the owner can update this timetable'.
2	Emily calls timetable.delete on 'SQA Week'.	FORBIDDEN 'Only the owner can delete this timetable'.

3	Student A renames the timetable via timetable.update to 'SQA Week 2' (101 characters is rejected).	Name changed; 101-character name rejected.
4	Student A: Timetable Settings > Delete Timetable and confirm.	Timetable is deleted with its 2 classes.
5	Student A and Emily open /timetable.	'SQA Week 2' is gone for both; neither has it as default; Emily no longer sees it under 'Shared with Me'.
6	Call timetable.delete with an unknown id.	NOT_FOUND 'Timetable not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-TT-02: Default timetable [Version : 1] REQ-TT-14: Update and delete timetable (owner only) [Version : 1]	
Keywords:	None	

3.Test Suite : Parwar - AI Assistant, Quizzes, Study Plans & Collaborative Notes (Iterations 3-4)

Tester: Parwar
Software under test: UniShare (github.com/Seganation/unishare)
Test basis: UniShare_Iteration_3.md AI Learning Assistant use cases (UC2 Chat with AI Assistant, UC-S3-011..UC-S3-020, Sequences 2-4, Use Case 3 Generate AI Quiz) and UniShare_Iteration_4.md UC-S4-002 Real-Time Collaborative Notes; code: src/server/api/routers/ai.ts, src/server/ai/*, src/app/api/chat, src/app/api/ai/generate-note, src/app/api/notes, src/app/api/liveblocks-auth, src/components/ai/*, src/components/notes/*, src/app/(student)/ai, src/app/(student)/courses/[id]/notes.
Techniques: Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table (DT), State Transition (ST), Use Case / scenario testing (UC), Error Guessing (EG).

Common test data (prisma/seed.ts, every seeded password is admin123)
Student A - Sarah Ahmed (sarah@student.utm.my), APPROVED, course owner
Student B - John Doe (john@student.utm.my), APPROVED, ACCEPTED CONTRIBUTOR on the test course
Student C - David Kim (david@student.utm.my), APPROVED, ACCEPTED VIEWER on the test course
Student D - Michael Chen (michael@student.utm.my), APPROVED, no relation to the test course
Student E - Alice Wong (alice@student.utm.my), role PENDING (not approved)
Course: "SECJ3343 Software Quality Assurance" created by Student A (default cards incl. Notes), description "Testing techniques and quality models".
AI provider: Google Gemini (GOOGLE_GENERATIVE_AI_API_KEY set, chat badge 'Gemini 2.5 Flash'); Liveblocks configured (LIVEBLOCKS_SECRET_KEY). API calls are made from the browser session of the named user (tRPC /api/trpc/ai.*, REST /api/chat, /api/ai/generate-note, /api/notes, /api/liveblocks-auth).

3.1.Test Suite : TS-P1 Chat with AI Assistant (UC-S3-012)

/ai chat page. Techniques: UC, EP, BVA on message length (0, 1, 5000, 5001), EG.

Test Case US-42: Start a new AI conversation (main flow) [Version : 1]		
Author:	parwar	
Summary: Start a new AI conversation (main flow). **Test design technique:** Use case (main flow), EP valid **Requirements:** REQ-P-CHAT-03, REQ-P-CHAT-04		
Preconditions: Student A logged in; AI provider reachable; Student A has 0 conversations.		
#:	Step actions:	Expected Results:
1	Open AI Assistant (/ai).	Empty chat with suggestions, input 'Ask AI anything...', badge 'Gemini 2.5 Flash', sidebar button 'New Chat'.
2	Type 'Explain the difference between let and const in JavaScript' and press Enter.	Message shown as user message; loader shown and Send disabled while the answer streams.
3	Wait until the answer is complete.	Full AI answer shown; URL is now /ai/<conversation id>.
4	Look at the sidebar.	New conversation on top with a short AI title (max 6 words, e.g. 'JavaScript let vs const'); footer 'Showing 1 of 1'.
5	Reload the page.	Both messages are still shown (saved); assistant message has tokensUsed > 0 in the database.
Execution type:	Manual	

Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-CHAT-03: New conversation is created on first message [Version : 1] REQ-P-CHAT-04: Streaming response and persistence [Version : 1]
Keywords:	None

Test Case US-43: Empty and whitespace-only message is not sent [Version : 1]		
Author:	parwar	
Summary: Empty and whitespace-only message is not sent. **Test design technique:** EP invalid class (empty), BVA (length 0) **Requirements:** REQ-P-CHAT-02		
Preconditions: Student A on /ai with a new chat.		
#:	Step actions:	Expected Results:
1	Leave the input empty.	Send button disabled.
2	Type 5 spaces and press Enter.	No message is sent and no conversation is created (Iteration 3 A1: 'Message cannot be empty').
3	Type 'a' (1 character) and send.	Message sent and answered (lower valid boundary).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-CHAT-02: Message validation [Version : 1]	
Keywords:	None	

Test Case US-44: Message length boundaries (4999, 5000, 5001 characters) [Version : 1]		
Author:	parwar	
Summary: Message length boundaries (4999, 5000, 5001 characters). **Test design technique:** BVA **Requirements:** REQ-P-CHAT-02		
Preconditions: Student A on /ai, new chat.		
#:	Step actions:	Expected Results:
1	Send a message of 4999 x 'a'.	Accepted and answered.
2	Send a message of exactly 5000 characters.	Accepted and answered.
3	Send a message of 5001 characters.	Rejected with 'Message must be under 5000 characters' and character count; nothing sent.

4	Call ai.sendMessage on an existing conversation with content of 10000 and 10001 characters.	10000 accepted, 10001 rejected by validation (implementation limit).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-CHAT-02: Message validation [Version : 1]	
Keywords:	None	

Test Case US-45: Continue a previous conversation with context [Version : 1]		
Author:	parwar	
Summary:		
Continue a previous conversation with context. **Test design technique:** Use case (UC-S3-013), ST (existing conversation) **Requirements:** REQ-P-CHAT-05, REQ-P-CHAT-04		
Preconditions:		
Student A has conversation 'JavaScript let vs const' with 2 messages.		
#:	Step actions:	Expected Results:
1	Click the conversation in the sidebar.	URL /ai/<id>; the 2 earlier messages shown oldest first.
2	Send 'Give me a code example of the second one'.	Answer refers to const (uses earlier context).
3	Reload.	Conversation now has 4 messages; it is first in the sidebar (latest updated).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-CHAT-04: Streaming response and persistence [Version : 1] REQ-P-CHAT-05: Continue a previous conversation [Version : 1]	
Keywords:	None	

Test Case US-46: Chat access without approval, session or conversation id [Version : 1]		
Author:	parwar	
Summary:		
Chat access without approval, session or conversation id. **Test design technique:** EP (APPROVED / PENDING / logged out), EG (missing id) **Requirements:** REQ-P-CHAT-01, REQ-P-CHAT-03		
Preconditions:		
Logged out browser; Student A (APPROVED) and Student E (PENDING) accounts available.		
#:	Step actions:	Expected Results:
1	Logged out, POST /api/chat with a message body.	HTTP 401 'Unauthorized'.
2	Logged out, open /ai.	Redirected to the login page.

3	Log in as Student A and POST /api/chat with a message but no conversationId.	HTTP 400 'No conversation ID provided'; no conversation created.
4	Log in as Student E (PENDING) and open /ai.	Redirected to /waiting-approval; chat not shown.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-CHAT-01: Only approved, logged-in students use the AI assistant [Version : 1] REQ-P-CHAT-03: New conversation is created on first message [Version : 1]	
Keywords:	None	

Test Case US-47: AI server unavailable [Version : 1]		
Author:	parwar	
Summary: AI server unavailable. **Test design technique:** EG (external system down) **Requirements:** REQ-P-CHAT-09		
Preconditions: Student A on /ai. AI provider blocked (invalid GOOGLE_GENERATIVE_AI_API_KEY or network off).		
#:	Step actions:	Expected Results:
1	Send 'Hello'.	Red alert shown: 'Unable to connect to AI server. Please make sure Ollama is running.' or the server error; page does not crash.
2	Restore the AI provider and send 'Hello' again.	Answer received; student could retry.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-CHAT-09: AI unavailable handling [Version : 1]	
Keywords:	None	

3.2.Test Suite : TS-P2 Conversation History & Privacy (UC-S3-019/020)

Sidebar history, paging, delete, ownership. Techniques: BVA (5/6, 15/16 conversations), ST, DT.

Test Case US-48: Sidebar paging boundaries (5, 6 and 16 conversations) [Version : 1]		
Author:	parwar	
Summary: Sidebar paging boundaries (5, 6 and 16 conversations). **Test design technique:** BVA (page size 5, step 10) **Requirements:** REQ-P-CHAT-06		
Preconditions: Student A has exactly 5 conversations.		
#:	Step actions:	Expected Results:

1	Open /ai.	5 conversations, newest updated first; 'Showing 5 of 5'; no 'Load More'.
2	Create a 6th conversation and reload.	'Showing 5 of 6' and 'Load More' button.
3	Click 'Load More'.	'Showing 6 of 6'; 'Load More' disappears.
4	With 16 conversations reload and click 'Load More' once.	'Showing 15 of 16'; Load More again gives 'Showing 16 of 16'.
5	Call ai.getConversations with limit 0, 1, 100, 101.	0 and 101 rejected; 1 and 100 accepted.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-CHAT-06: Conversation history sidebar [Version : 1]	
Keywords:	None	

Test Case US-49: New Chat and untitled conversation [Version : 1]		
Author:	parwar	
Summary: New Chat and untitled conversation. **Test design technique:** EP (titled / untitled) **Requirements:** REQ-P-CHAT-06, REQ-P-CHAT-03		
Preconditions: Student A inside conversation /ai/<id>.		
#:	Step actions:	Expected Results:
1	Click 'New Chat'.	URL /ai; empty chat; previous conversation still in sidebar.
2	Call ai.createConversation with title '' (empty) via API, reload.	Conversation listed as 'Untitled Chat'.
3	Call ai.createConversation without title.	Conversation titled 'New Conversation', temperature 0.7.
4	Call ai.createConversation with temperature -0.1, 0, 2, 2.1.	-0.1 and 2.1 rejected; 0 and 2 accepted.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Low	
Requirements	REQ-P-CHAT-03: New conversation is created on first message [Version : 1] REQ-P-CHAT-06: Conversation history sidebar [Version : 1]	
Keywords:	None	

Test Case US-50: Delete conversation - confirm and cancel [Version : 1]	
Author:	parwar
Summary:	

Delete conversation - confirm and cancel.

Test design technique: ST (exists -> deleted), DT (OK / Cancel)

Requirements: REQ-P-CHAT-07

Preconditions:

Student A has conversation X with 4 messages, currently open.

#:	Step actions:	Expected Results:
1	Hover X in the sidebar, click the delete icon, choose Cancel in 'Delete this conversation? This action cannot be undone.'	Nothing deleted; X still listed.
2	Repeat and choose OK.	X removed from the sidebar; chat returns to a new empty chat at /ai.
3	Open /ai/<X id>.	No messages of X are shown (conversation and its 4 messages deleted).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-CHAT-07: Delete a conversation [Version : 1]
Keywords:	None

Test Case US-51: User cannot read, continue or delete another user's conversation [Version : 1]

Author:	parwar

Summary:

User cannot read, continue or delete another user's conversation.

Test design technique: DT (actor = other user x action), security negative

Requirements: REQ-P-CHAT-08

Preconditions:

Student A owns conversation X (id known). Student B logged in.

#:	Step actions:	Expected Results:
1	As Student B look at the sidebar.	X is not listed.
2	As Student B call ai.getConversation with X's id.	NOT_FOUND 'Conversation not found'.
3	As Student B call ai.sendMessage on X with 'hi'.	NOT_FOUND 'Conversation not found'; no message added to X.
4	As Student B call ai.deleteConversation with X's id.	NOT_FOUND 'Conversation not found'; X still exists for Student A.
5	As Student B POST /api/chat with conversationId = X's id.	Student A's messages are not returned or changed (must not create or reuse X for Student B).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-CHAT-08: Conversation privacy [Version : 1]

Keywords:	None

3.3.Test Suite : TS-P3 Generate AI Quiz (UC-S3-014)

Course > Quizzes > Generate Quiz. Techniques: UC, BVA (questions 0/1/5/50/51), EP (difficulty, types), DT (access).

Test Case US-52: Generate a quiz for own course (main flow) [Version : 1]		
Author:	parwar	
Summary: Generate a quiz for own course (main flow). **Test design technique:** Use case main flow, EP valid **Requirements:** REQ-P-QUIZ-01, REQ-P-QUIZ-04		
Preconditions: Student A logged in, owns SECJ3343; AI provider reachable.		
#:	Step actions:	Expected Results:
1	Open SECJ3343 > tab Quizzes.	'Course Quizzes' with 'Generate Quiz' button; list shows 'No quizzes yet'.
2	Click Generate Quiz.	Form: Topic *, Number of Questions = 10 (slider 5-50), Difficulty = Medium, Multiple Choice and True/False checked, Short Answer unchecked.
3	Topic 'Boundary Value Analysis', 10 questions, Medium; click Generate Quiz.	Button shows 'Generating...', fields disabled.
4	Wait for completion.	Toast 'Quiz generated successfully!'; quiz taker opens with '0 of 10 answered' and badge '10 questions'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-QUIZ-01: Quiz generation form and limits [Version : 1] REQ-P-QUIZ-04: Successful generation is saved [Version : 1]	
Keywords:	None	

Test Case US-53: Topic is required [Version : 1]		
Author:	parwar	
Summary: Topic is required. **Test design technique:** EP invalid (empty / whitespace), BVA (length 0/1) **Requirements:** REQ-P-QUIZ-01		
Preconditions: Student A on Generate Quiz form of SECJ3343.		
#:	Step actions:	Expected Results:
1	Leave Topic empty and click Generate Quiz.	Toast 'Please enter a topic'; no request sent.
2	Topic = 3 spaces, Generate Quiz.	Toast 'Please enter a topic'.

3	Topic = 'X' (1 character), Generate Quiz.	Quiz generated (lower boundary valid).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-QUIZ-01: Quiz generation form and limits [Version : 1]	
Keywords:	None	

Test Case US-54: No question type selected [Version : 1]		
Author:	parwar	
Summary: No question type selected. **Test design technique:** DT (MC x TF x SA all unchecked) **Requirements:** REQ-P-QUIZ-02		
Preconditions: Student A on Generate Quiz form, topic 'Equivalence Partitioning'.		
#:	Step actions:	Expected Results:
1	Uncheck Multiple Choice and True/False (Short Answer unchecked), click Generate Quiz.	Toast 'Please select at least one question type'; no quiz generated.
2	Check only Short Answer, Generate Quiz.	Quiz generated with SHORT_ANSWER questions only.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-QUIZ-02: At least one question type [Version : 1]	
Keywords:	None	

Test Case US-55: Question count boundaries [Version : 1]		
Author:	parwar	
Summary: Question count boundaries. **Test design technique:** BVA (UI 5..50 step 5, API 1..50) **Requirements:** REQ-P-QUIZ-01		
Preconditions: Student A on Generate Quiz form, topic 'Decision Tables'.		
#:	Step actions:	Expected Results:
1	Drag slider to the far left.	Value shows 5 (cannot go lower; labels '5 questions' / '50 questions').
2	Generate with 5.	Quiz with 5 questions.
3	Drag slider to the far right and generate.	Value 50; quiz with 50 questions.

4	Call ai.generateQuiz with questionCount 0, 1, 50, 51.	0 and 51 rejected by validation; 1 and 50 accepted.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-QUIZ-01: Quiz generation form and limits [Version : 1]	
Keywords:	None	

Test Case US-56: Difficulty and question type partitions [Version : 1]		
Author:	parwar	
Summary:		
Difficulty and question type partitions. **Test design technique:** EP (easy / medium / hard; MC / TF / SA) **Requirements:** REQ-P-QUIZ-01, REQ-P-QUIZ-05		
Preconditions:		
Student A on Generate Quiz form.		
#:	Step actions:	Expected Results:
1	Generate 'SQL Joins', Easy, only True/False.	All questions show the options True and False.
2	Generate 'SQL Joins', Hard, only Multiple Choice.	All questions show radio options from the AI.
3	Call ai.generateQuiz with difficulty 'expert'.	Rejected by validation (only easy, medium, hard).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Low	
Requirements	REQ-P-QUIZ-01: Quiz generation form and limits [Version : 1] REQ-P-QUIZ-05: Taking a quiz [Version : 1]	
Keywords:	None	

Test Case US-57: Quiz generation access by course relation [Version : 1]		
Author:	parwar	
Summary:		
Quiz generation access by course relation. **Test design technique:** DT (owner / contributor / viewer / unrelated / unknown course) **Requirements:** REQ-P-QUIZ-03		
Preconditions:		
SECJ3343 owned by Student A; B CONTRIBUTOR, C VIEWER (both ACCEPTED); D unrelated.		
#:	Step actions:	Expected Results:
1	As Student B generate a quiz in SECJ3343.	Generated (accepted collaborator).
2	As Student C generate a quiz in SECJ3343.	Generated (viewer is an accepted collaborator).

3	As Student D call ai.generateQuiz with courseId of SECJ3343.	FORBIDDEN 'You don't have access to this course'; nothing saved.
4	As Student A call ai.generateQuiz with courseId 'does-not-exist'.	NOT_FOUND 'Course not found'.
5	As Student D call ai.generateQuiz with noteId of a SECJ3343 note; then with noteId 'nope'.	'You don't have access to this note'; then 'Note not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-QUIZ-03: Course / note context access [Version : 1]	
Keywords:	None	

Test Case US-58: Invalid AI response is not saved [Version : 1]		
Author:	parwar	
Summary: Invalid AI response is not saved. **Test design technique:** EG (malformed JSON from AI) **Requirements:** REQ-P-QUIZ-04		
Preconditions: Student A on Generate Quiz form; AI stubbed to return plain text instead of JSON.		
#:	Step actions:	Expected Results:
1	Generate a quiz on 'Mutation testing'.	Toast 'Failed to generate quiz. The AI response was not in the expected format.'; form enabled again.
2	Open the Quizzes list.	No new quiz was added.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-QUIZ-04: Successful generation is saved [Version : 1]	
Keywords:	None	

3.4.Test Suite : TS-P4 Attempt Quiz & Grading (UC-S3-015)

Quiz taker and results. Techniques: ST (not started -> answering -> submitted), BVA on score bands, EP on answer matching.

Test Case US-59: Submit is enabled only when all questions are answered [Version : 1]		
Author:	parwar	
Summary: Submit is enabled only when all questions are answered. **Test design technique:** ST, BVA (answered n-1 / n) **Requirements:** REQ-P-QUIZ-05		
Preconditions: Student A opens a 5-question quiz 'BVA Quiz'.		
#:	Step actions:	Expected Results:

1	Look at the header and navigation.	'0 of 5 answered', '0% complete'; Previous disabled on question 1.
2	Answer questions 1-4 using Next.	'4 of 5 answered', '80% complete'; answered buttons show a check icon.
3	Go to question 5 without answering.	'Submit Quiz' disabled and text 'Please answer all questions before submitting'.
4	Answer question 5.	'5 of 5 answered', '100% complete'; Submit Quiz enabled.
5	Click Submit Quiz.	Button shows 'Submitting...', then the results page.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-QUIZ-05: Taking a quiz [Version : 1]	
Keywords:	None	

Test Case US-60: Score bands on the results page [Version : 1]		
Author:	parwar	
Summary: Score bands on the results page. **Test design technique:** BVA (scores 100, 90, 80, 70, 60, 50) **Requirements:** REQ-P-QUIZ-06		
Preconditions: Student A has a 10-question multiple-choice quiz with known correct answers.		
#:	Step actions:	Expected Results:
1	Submit with 10 correct.	Score 100%, 'Excellent!'.
2	Retake: 9 correct.	90%, 'Excellent!' (boundary 90).
3	Retake: 8 correct.	80%, 'Great job!'.
4	Retake: 7 correct.	70%, 'Good work!'.
5	Retake: 6 correct.	60%, 'Not bad!'.
6	Retake: 5 correct.	50%, 'Keep studying!'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-QUIZ-06: Automatic grading and score [Version : 1]	
Keywords:	None	

Test Case US-61: Answer matching rules per question type [Version : 1]	
Author:	parwar

Summary: Answer matching rules per question type. **Test design technique:** EP (exact / different case / extra spaces) x type **Requirements:** REQ-P-QUIZ-06		
Preconditions: Quiz with Q1 SHORT_ANSWER correct 'Boundary Value Analysis', Q2 TRUE_FALSE correct 'True', Q3 MULTIPLE_CHOICE correct option 'B. Black-box'.		
#:	Step actions:	Expected Results:
1	Answer Q1 ' boundary value analysis ', Q2 True, Q3 option 'B. Black-box'; submit.	All 3 correct (short answer trimmed and case-insensitive); score 100%.
2	Retake: Q1 'BVA', others correct.	Q1 incorrect; score 66.67% shown rounded, not 66.666...%.
3	Retake on a quiz where the stored True/False answer is 'true' (lower case) and choose True.	Answer counted correct (True/False must not fail on letter case).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-QUIZ-06: Automatic grading and score [Version : 1]	
Keywords:	None	

Test Case US-62: Attempt is saved, best score and retake [Version : 1]	
Author:	parwar
Summary: Attempt is saved, best score and retake. **Test design technique:** ST (attempt 1 -> attempt 2), use case **Requirements:** REQ-P-QUIZ-07, REQ-P-QUIZ-06	
Preconditions: Student A has quiz 'BVA Quiz' with one attempt of 60%.	

#:	Step actions:	Expected Results:
1	On the results page click 'Retake Quiz' and score 80%.	New attempt saved with completion time; results 80%.
2	Go back to the Quizzes list.	'BVA Quiz' shows 'Best Score: 80%'; newest quiz listed first.
3	Open the attempts of 'BVA Quiz'.	Two attempts listed (80% newest, 60%).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-P-QUIZ-06: Automatic grading and score [Version : 1] REQ-P-QUIZ-07: Quiz list and repeated attempts [Version : 1]
Keywords:	None

Test Case US-63: Another user's quiz cannot be opened, attempted or deleted [Version : 1]

Author:	parwar	
Summary: Another user's quiz cannot be opened, attempted or deleted. **Test design technique:** DT (actor = non-owner), security negative **Requirements:** REQ-P-QUIZ-08		
Preconditions: Student A owns quiz Q in SECJ3343; Student B (contributor) logged in.		
#:	Step actions:	Expected Results:
1	As Student B open SECJ3343 > Quizzes.	Q is not listed (list shows only B's quizzes).
2	As Student B call ai.getQuiz with Q's id.	NOT_FOUND 'Quiz not found'.
3	As Student B call ai.submitQuizAttempt for Q.	NOT_FOUND 'Quiz not found'; no attempt stored.
4	As Student B call ai.deleteQuiz for Q.	NOT_FOUND 'Quiz not found'; Q still exists.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-QUIZ-08: Quiz ownership and delete [Version : 1]	
Keywords:	None	

Test Case US-64: Delete own quiz [Version : 1]		
Author:	parwar	
Summary: Delete own quiz. **Test design technique:** ST (exists -> deleted), DT (OK / Cancel) **Requirements:** REQ-P-QUIZ-08, REQ-P-QUIZ-07		
Preconditions: Student A has exactly one quiz 'BVA Quiz' with 2 attempts.		
#:	Step actions:	Expected Results:
1	Click delete on 'BVA Quiz' and Cancel 'Are you sure you want to delete this quiz? This action cannot be undone.'	Quiz still listed.
2	Delete again and confirm.	Quiz removed; list shows 'No quizzes yet'.
3	Call ai.getQuiz with the old id.	'Quiz not found' (questions and attempts deleted too).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-QUIZ-07: Quiz list and repeated attempts [Version : 1] REQ-P-QUIZ-08: Quiz ownership and delete [Version : 1]	
Keywords:	None	

3.5.Test Suite : TS-P5 AI Study Plans (UC-S3-016/017)

Course > Study Plans. Techniques: UC, BVA (weeks 0/1/16/17, hours 0/1/40/41), ST (task open <-> done), DT.

Test Case US-65: Generate a study plan (main flow) [Version : 1]		
Author:	parwar	
Summary: Generate a study plan (main flow). **Test design technique:** Use case main flow, EP valid **Requirements:** REQ-P-PLAN-01, REQ-P-PLAN-03		
Preconditions: Student A owns SECJ3343; AI reachable; no study plans yet.		
#:	Step actions:	Expected Results:
1	Open SECJ3343 > Study Plans.	'Study Plans' header, 'Generate Plan' button, 'No study plans yet'.
2	Click Generate Plan.	Course preselected; Number of Weeks 4, Hours Per Week 5, Learning Goal (Optional), Target Deadline (Optional).
3	Weeks 4, Hours 10, goal 'Prepare for final exam', deadline today + 30 days; Generate.	Toast 'Study plan generated successfully!'; plan opens.
4	Check the plan.	4 weeks in order Week 1..4 each with tasks; 'Overall Progress' 0%; endDate = deadline, startDate = deadline - 28 days.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-PLAN-01: Study plan form and limits [Version : 1] REQ-P-PLAN-03: Generated plan is saved with weeks and tasks [Version : 1]	
Keywords:	None	

Test Case US-66: Weeks and hours boundaries [Version : 1]		
Author:	parwar	
Summary: Weeks and hours boundaries. **Test design technique:** BVA (weeks 0/1/16/17, hours 0/1/40/41) **Requirements:** REQ-P-PLAN-01		
Preconditions: Student A on Generate Study Plan for SECJ3343.		
#:	Step actions:	Expected Results:
1	Move Number of Weeks slider fully left, then fully right.	Minimum 1, maximum 16.
2	Move Hours Per Week slider fully left, then fully right.	Minimum 1, maximum 40.
3	Generate with weeks 1, hours 1.	Plan with exactly 1 week.

4	Generate with weeks 16, hours 40.	Plan with 16 weeks (Iteration 3 says max 12 - record deviation).
5	Call ai.generateStudyPlan with weekCount 0 and 17, hoursPerWeek 0 and 41.	All four rejected by validation.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-PLAN-01: Study plan form and limits [Version : 1]	
Keywords:	None	

Test Case US-67: Course is required and deadline cannot be in the past [Version : 1]		
Author:	parwar	
Summary: Course is required and deadline cannot be in the past. **Test design technique:** EP invalid (no course, past date), EG **Requirements:** REQ-P-PLAN-01, REQ-P-PLAN-03		
Preconditions: Student A opens Generate Study Plan from AI pages without a preselected course.		
#:	Step actions:	Expected Results:
1	Leave course empty.	Generate button disabled; a forced submit shows 'Please select a course'.
2	Open the deadline calendar.	Yesterday and earlier dates are disabled; today+1 can be picked.
3	Select SECJ3343, no goal, no deadline; Generate.	Plan created; stored prompt 'Study plan for course'; no start or end date.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-PLAN-01: Study plan form and limits [Version : 1] REQ-P-PLAN-03: Generated plan is saved with weeks and tasks [Version : 1]	
Keywords:	None	

Test Case US-68: Study plan access by course relation [Version : 1]		
Author:	parwar	
Summary: Study plan access by course relation. **Test design technique:** DT (owner / collaborator / unrelated / unknown course) **Requirements:** REQ-P-PLAN-02		
Preconditions: SECJ3343 owned by Student A; Student B ACCEPTED contributor; Student D unrelated.		
#:	Step actions:	Expected Results:

1	As Student B generate a plan for SECJ3343.	Generated.
2	As Student D call ai.generateStudyPlan with SECJ3343 id.	FORBIDDEN 'You don't have access to this course'.
3	As Student A call ai.generateStudyPlan with courseId 'does-not-exist'.	NOT_FOUND 'Course not found'.
4	AI stub returns invalid JSON; generate as Student A.	'Failed to generate study plan. The AI response was not in the expected format.'; nothing saved.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-PLAN-02: Course access for study plans [Version : 1]	
Keywords:	None	

Test Case US-69: Track task completion and progress [Version : 1]

Author:	parwar	
Summary:		
Track task completion and progress. **Test design technique:** ST (open -> done -> open), BVA (0%, 100%) **Requirements:** REQ-P-PLAN-04		
Preconditions:		
Student A has plan P: 2 weeks, 4 tasks each (8 tasks), none completed.		
#:	Step actions:	Expected Results:
1	Open P.	Overall Progress 0%; each Week Progress 0%.
2	Tick 1 task in week 1.	Toast 'Task updated'; Overall 13% (1/8 = 12.5 rounded); Week 1 25%.
3	Tick the other 3 tasks of week 1.	Week 1 Progress 100% and shown as complete; Overall 50%.
4	Untick one task of week 1.	Week 1 75%; Overall 38%; task completedAt cleared.
5	Tick all 8 tasks.	Overall Progress 100%.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-PLAN-04: Track progress [Version : 1]	
Keywords:	None	

Test Case US-70: Plan ownership and delete [Version : 1]

Author:	parwar
Summary:	
Plan ownership and delete. **Test design technique:** DT (actor x action), ST (exists -> deleted)	

Requirements: REQ-P-PLAN-05

Preconditions:

Student A has plan P (task T id known). Student B logged in.

#:	Step actions:	Expected Results:
1	As Student B call ai.updateStudyPlanTask on T.	NOT_FOUND 'Task not found'; T unchanged.
2	As Student B call ai.deleteStudyPlan on P.	NOT_FOUND 'Study plan not found'.
3	As Student A delete P and confirm 'Are you sure you want to delete this study plan? This action cannot be undone.'	P removed; if it was the only plan, 'No study plans yet'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-P-PLAN-05: Plan ownership and delete [Version : 1]
Keywords:	None

Test Case US-71: Study plan survives course deletion [Version : 1]

Author:	parwar

Summary:

Study plan survives course deletion.

Test design technique: EG, EP (course deleted)

Requirements: REQ-P-PLAN-06

Preconditions:

Student A owns course 'Temp Course' with study plan P2.

#:	Step actions:	Expected Results:
1	Delete 'Temp Course'.	Course deleted.
2	Call ai.getStudyPlans without courseId.	P2 still returned with course = null (SET NULL as in Iteration 3); fails if P2 was cascade-deleted.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Low
Requirements	REQ-P-PLAN-06: Plan kept when the course is deleted [Version : 1]
Keywords:	None

3.6.Test Suite : TS-P6 Enhance Notes with AI (UC-S3-018)

POST /api/ai/generate-note. Techniques: EP (4 types + invalid), BVA (prompt 0/1/5000/5001), DT (role).

Test Case US-72: Each enhancement type returns content and is recorded [Version : 1]

Author:	parwar

Summary:

Each enhancement type returns content and is recorded.

Test design technique: EP (GENERATE / IMPROVE / SUMMARIZE / EXPAND)

Requirements: REQ-P-AIN-01, REQ-P-AIN-03

Preconditions:

Student A owns SECJ3343 with note page N 'Getting Started'.

#:	Step actions:	Expected Results:
1	POST /api/ai/generate-note {noteId: N, type: GENERATE, prompt: 'Black-box testing'}.	200 with success true, markdown generatedText, tokensUsed, generationId.
2	Repeat with IMPROVE, SUMMARIZE and EXPAND.	Each returns 200 and text matching the type.
3	Call ai.getNoteGenerations for N.	4 records newest first with types EXPAND, SUMMARIZE, IMPROVE, GENERATE.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-AIN-01: Enhancement request validation [Version : 1] REQ-P-AIN-03: Enhancement result and history [Version : 1]
Keywords:	None

Test Case US-73: Invalid enhancement requests [Version : 1]

Author:	parwar

Summary:

Invalid enhancement requests.

Test design technique: BVA (prompt 0/1/5000/5001), EP invalid type

Requirements: REQ-P-AIN-01

Preconditions:

Student A, note N.

#:	Step actions:	Expected Results:
1	prompt '' (0 chars).	400 'Invalid request'.
2	prompt 'a' (1) and prompt of 5000 chars.	Both 200.
3	prompt of 5001 chars.	400 'Invalid request'.
4	type 'TRANSLATE'.	400 'Invalid request'.
5	temperature 2.5.	400 'Invalid request'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-P-AIN-01: Enhancement request validation [Version : 1]
Keywords:	None

Test Case US-74: Enhancement permission by role [Version : 1]

Author:	parwar

Summary:

Enhancement permission by role.

Test design technique: DT (owner / contributor / viewer / unrelated / logged out / unknown note)

Requirements: REQ-P-AIN-02, REQ-P-AIN-03

Preconditions:

Note N in SECJ3343; B contributor, C viewer, D unrelated.

#:	Step actions:	Expected Results:
1	As Student B, GENERATE on N.	200.
2	As Student C, GENERATE on N.	403 'Forbidden: You don't have permission to edit this note'.
3	As Student D, GENERATE on N.	403 'Forbidden: You don't have permission to edit this note'.
4	Logged out, GENERATE on N.	401 'Unauthorized'.
5	As Student A, noteId 'missing'.	404 'Note not found'.
6	As Student B call ai.getNoteGenerations for N.	Only Student B's generation listed (not Student A's).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-AIN-02: Only owner or contributor can enhance a note [Version : 1] REQ-P-AIN-03: Enhancement result and history [Version : 1]
Keywords:	None

3.7.Test Suite : TS-P7 Collaborative Notes - Access & Roles (UC-S4-002)

Course > Notes card > /courses/<id>/notes. Techniques: DT (role x action), EP, EG.

Test Case US-75: Owner opens notes for the first time [Version : 1]

Author: parwar

Summary:

Owner opens notes for the first time.

Test design technique: Use case (UC-S4-002 steps 1-9), ST (no page -> first page)

Requirements: REQ-P-NOTE-01, REQ-P-NOTE-03

Preconditions:

Student A owns SECJ3343 which has no note pages.

#:	Step actions:	Expected Results:
1	Open SECJ3343 and click 'Open Collaborative Notes' on the Notes card.	Goes to /courses/<id>/notes and redirects to /courses/<id>/notes/<page id>.
2	Look at the page.	Page 'Getting Started' in sidebar 'Pages'; header 'SECJ3343 Software Quality Assurance', 'Collaborative Notes', badge OWNER.
3	Look at the editor.	Editable, 'All changes saved', 'Real-time collaboration active!' help text.

Execution type:	Manual

Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-NOTE-01: Open course notes [Version : 1] REQ-P-NOTE-03: Role-based editing (OWNER / CONTRIBUTOR / VIEWER) [Version : 1]
Keywords:	None

Test Case US-76: Viewer opens a course without pages [Version : 1]

Author:	parwar

Summary:

Viewer opens a course without pages.

Test design technique: EP (role VIEWER, no pages)

Requirements: REQ-P-NOTE-01

Preconditions:

Course 'Empty Notes Course' of Student A has no pages; Student C is ACCEPTED VIEWER.

#:	Step actions:	Expected Results:
1	As Student C open /courses/<id>/notes.	'No pages have been created yet. Ask a contributor or owner to create one.' and 'Back to Course'; no page created.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-P-NOTE-01: Open course notes [Version : 1]
Keywords:	None

Test Case US-77: Notes access by user relation [Version : 1]

Author:	parwar

Summary:

Notes access by user relation.

Test design technique: DT (owner / contributor / viewer / pending invite / unrelated / logged out / unknown course)

Requirements: REQ-P-NOTE-02

Preconditions:

SECJ3343 has pages; B CONTRIBUTOR, C VIEWER (ACCEPTED); Student D has a PENDING invitation only.

#:	Step actions:	Expected Results:
1	As Student B open the notes.	Opens, badge CONTRIBUTOR.
2	As Student C open the notes.	Opens, badge VIEWER.
3	As Student D open /courses/<id>/notes.	Redirected to /courses/<id>; notes not shown.
4	Logged out open the notes URL.	Redirected to /login.
5	As Student A open /courses/unknown-id/notes.	Redirected to /courses.

Execution type:	Manual
Estimated exec. duration (min):	

Importance:	High
Requirements	REQ-P-NOTE-02: Only course members can open notes [Version : 1]
Keywords:	None

Test Case US-78: Viewer is read-only (UI and server) [Version : 1]		
Author:	parwar	
Summary:		
Viewer is read-only (UI and server). **Test design technique:** DT (role = VIEWER column), security negative **Requirements:** REQ-P-NOTE-03, REQ-P-NOTE-07		
Preconditions:		
Student C is ACCEPTED VIEWER on SECJ3343; page 'Getting Started' exists.		
#:	Step actions:	Expected Results:
1	As Student C open the page.	Banner with eye icon 'Viewing in Read-Only Mode' ('You're a Viewer on this course...'); badge VIEWER; no '+' and no page menu.
2	Try to type in the editor and in the title.	Nothing can be typed; title field disabled.
3	POST /api/notes/create {courseId}.	403 'You don't have permission to create pages'.
4	PATCH /api/notes/<page id> {title: 'Hacked'}.	403 'You don't have permission to edit this page'.
5	DELETE /api/notes/<page id>.	403 'You don't have permission to delete this page'.
6	With the Liveblocks token from /api/liveblocks-auth, send a document change to the room.	Change rejected (read-only access); owner does not see it. Fails if token grants full access.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-P-NOTE-03: Role-based editing (OWNER / CONTRIBUTOR / VIEWER) [Version : 1] REQ-P-NOTE-07: Delete pages and server-side permission checks [Version : 1]	
Keywords:	None	

Test Case US-79: Liveblocks authentication endpoint [Version : 1]		
Author:	parwar	
Summary:		
Liveblocks authentication endpoint. **Test design technique:** EP / EG on /api/liveblocks-auth inputs **Requirements:** REQ-P-NOTE-02		
Preconditions:		
Page id P of SECJ3343 known.		
#:	Step actions:	Expected Results:
1	Logged out POST {room: P}.	401 'Unauthorized'.

2	As Student A POST {} (no room).	400 'Bad Request - Room ID missing'.
3	As Student A POST body 'not json'.	400 'Bad Request - Invalid JSON'.
4	As Student A POST {room: 'nope'}.	404 'Page not found'.
5	As Student D (unrelated) POST {room: P}.	403 'Forbidden'.
6	As Student B POST {room: P}.	200 with token; userInfo.role CONTRIBUTOR.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-P-NOTE-02: Only course members can open notes [Version : 1]	
Keywords:	None	

Test Case US-80: Unknown or other course page id [Version : 1]		
Author:	parwar	
Summary:		
Unknown or other course page id. **Test design technique:** EG, EP invalid page id **Requirements:** REQ-P-NOTE-01		
Preconditions:		
SECJ3343 first page P1; course 'Other' (also owned by A) with page Q.		
#:	Step actions:	Expected Results:
1	As Student A open /courses/<SECJ3343>/notes.	Redirected to P1.
2	Open /courses/<SECJ3343>/notes/does-not-exist.	Redirected to P1.
3	Open /courses/<SECJ3343>/notes/<Q>.	Redirected to P1 (page of another course not shown).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Low	
Requirements	REQ-P-NOTE-01: Open course notes [Version : 1]	
Keywords:	None	

3.8.Test Suite : TS-P8 Collaborative Notes - Real-Time Editing & Pages

Two browsers (A and B) on the same course. Techniques: UC, ST (online -> offline -> online), BVA (5/6 others), EG.

Test Case US-81: Two students edit the same page in real time [Version : 1]	
Author:	parwar
Summary:	
Two students edit the same page in real time. **Test design technique:** Use case (UC-S4-002 steps 10-27) **Requirements:** REQ-P-NOTE-04, REQ-P-NOTE-05	

Preconditions:

Student A (Chrome) and Student B (Firefox, CONTRIBUTOR) both on page 'Getting Started'.

#:	Step actions:	Expected Results:
1	A types heading 'Chapter 1: Introduction'.	B sees the text appear without reloading.
2	B clicks below and types 'Key Concepts:' and a bullet list.	A sees B's text; both documents identical.
3	Look at the Active Collaborators bar in both browsers.	Each shows own avatar ('<name> (You)') and the other user; '2 online'.
4	Both reload.	All text still present; 'All changes saved'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-NOTE-04: Real-time synchronisation and persistence [Version : 1] REQ-P-NOTE-05: Presence of live collaborators [Version : 1]
Keywords:	None

Test Case US-82: Simultaneous edit of the same text converges [Version : 1]

Author:	parwar

Summary:

Simultaneous edit of the same text converges.

Test design technique: EG (concurrency), Iteration 4 A2

Requirements: REQ-P-NOTE-04

Preconditions:

A and B on the same page containing the word 'Testing'.

#:	Step actions:	Expected Results:
1	At the same time A types 'Unit ' before and B types ' Basics' after 'Testing'.	Both editors end with the same text (e.g. 'Unit Testing Basics'); no error, no lost characters.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-P-NOTE-04: Real-time synchronisation and persistence [Version : 1]
Keywords:	None

Test Case US-83: Presence list boundaries and leaving [Version : 1]

Author:	parwar

Summary:

Presence list boundaries and leaving.

Test design technique: BVA (5 / 6 other users), ST (online -> left)

Requirements: REQ-P-NOTE-05

Preconditions:

Student A on a page; 6 other course members available (owner/collaborator accounts).

#:	Step actions:	Expected Results:
1	5 others join.	5 other avatars shown, no '+'; '6 online'.
2	6th other joins.	5 avatars and '+1'; '7 online'.
3	Student B closes the browser.	B's avatar removed from A's bar; count decreases by 1.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Low
Requirements	REQ-P-NOTE-05: Presence of live collaborators [Version : 1]
Keywords:	None

Test Case US-84: Create page and nested sub-page [Version : 1]

Author:	parwar

Summary:

Create page and nested sub-page.

Test design technique: Use case (steps 28-38), ST

Requirements: REQ-P-NOTE-06

Preconditions:

Student A on SECJ3343 notes with page 'Getting Started'; Student B on the same course.

#:	Step actions:	Expected Results:
1	A clicks '+' next to 'Pages'.	Page 'Untitled' created after 'Getting Started' and opened.
2	A changes the title to 'Week 1 Notes' and presses Enter.	'Saving title...' then sidebar shows 'Week 1 Notes'.
3	A opens the menu of 'Week 1 Notes' and clicks 'Add sub-page'.	Child page 'Untitled' shown indented under 'Week 1 Notes'.
4	Collapse 'Week 1 Notes'.	Sub-page hidden; expand shows it again.
5	B refreshes the sidebar.	B sees 'Week 1 Notes' and its sub-page.
6	A types in the sub-page while B stays on 'Getting Started'.	B's page does not change (separate room).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-P-NOTE-06: Create pages and nested sub-pages [Version : 1]
Keywords:	None

Test Case US-85: Delete a page with sub-pages [Version : 1]

Author:	parwar

Summary:

Delete a page with sub-pages.

Test design technique: ST (page -> deleted with children), DT (OK / Cancel)

Requirements: REQ-P-NOTE-07

Preconditions:

Student B (CONTRIBUTOR) on 'Week 1 Notes' which has one sub-page; 'Getting Started' is first page.

#:	Step actions:	Expected Results:
1	Delete 'Week 1 Notes' and Cancel 'Are you sure you want to delete this page?'	Nothing deleted.
2	Delete again and confirm.	'Week 1 Notes' and its sub-page removed; 'Getting Started' opens.
3	Open the old sub-page URL.	Redirected to 'Getting Started' (sub-page deleted by cascade).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-P-NOTE-07: Delete pages and server-side permission checks [Version : 1]
Keywords:	None

Test Case US-86: Offline edits sync after reconnect [Version : 1]

Author:	parwar

Summary:

Offline edits sync after reconnect.

Test design technique: ST (online -> offline -> online), Iteration 4 A1

Requirements: REQ-P-NOTE-08, REQ-P-NOTE-04

Preconditions:

A and B on the same page.

#:	Step actions:	Expected Results:
1	B goes offline (DevTools > Network > Offline) and types 'offline line'.	B can keep typing; A does not see it yet.
2	A types 'online line' meanwhile.	A's text saved.
3	B goes back online.	Both see 'offline line' and 'online line'; documents identical.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-P-NOTE-04: Real-time synchronisation and persistence [Version : 1] REQ-P-NOTE-08: Network interruption [Version : 1]
Keywords:	None

4.Test Suite : Rawa Dara - Sharing, Articles & Notifications (Iteration 4)

Tester: Rawa Dara
Software under test: UniShare (github.com/Seganation/unishare)
Test basis: UniShare_Iteration_4.md use cases UC-S4-001, UC-S4-003, UC-S4-004 and the tRPC routers course.ts, resource.ts, article.ts, notification.ts.
Techniques: Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table (DT), State Transition (ST), Use Case / scenario testing (UC), Error Guessing (EG).

Common test data
Student A - Sarah Ahmed (sarah@student.utm.my), course owner, APPROVED, Faculty of Computing
Student B - John Doe (john@student.utm.my), APPROVED, Faculty of Computing (same faculty as A)
Student C - Michael Chen (michael@student.utm.my), APPROVED, Faculty of Engineering (different faculty)
Student D - Alice Wong (alice@student.utm.my), role PENDING (not approved)
Seed data: prisma/seed.ts (all seeded passwords documented there).
Course: "SECJ3343 Software Quality Assurance" created by Student A with one resource card "Lecture Slides" (file uploads allowed) holding file week1.pdf.

4.1.Test Suite : TS-R1 Share Course (UC-S4-001)

Owner invites students to a course. Techniques: EP, DT, EG.

Test Case US-1: Owner invites same-faculty student as VIEWER [Version : 1]		
Author:	rawa	
Summary:		
Owner invites same-faculty student as VIEWER. **Test design technique:** Use case (main flow), EP valid class **Requirements:** REQ-SHR-03, REQ-SHR-07, REQ-NOT-01		
Preconditions:		
Student A logged in and owns course SECJ3343. Student B exists, same faculty, not yet invited.		
#:	Step actions:	Expected Results:
1	Open Courses > SECJ3343 Software Quality Assurance > Collaborators.	Collaborators page shows 'Manage who has access to SECJ3343 Software Quality Assurance'.
2	Enter Student B's email in the invite field (colleague@university.edu placeholder).	Email accepted in the field.
3	Select permission 'Viewer' and click Invite.	Success message shown; Student B listed with role Viewer and status Pending.
4	Log in as Student B and open the notification bell.	Notification 'New course invitation': 'Student A invited you to collaborate on SECJ3343 Software Quality Assurance'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-SHR-03: Invitation carries a permission level [Version : 1] REQ-SHR-07: Owner manages collaborators [Version : 1] REQ-NOT-01: Event notifications [Version : 1]	
Keywords:	None	

Test Case US-2: Owner invites same-faculty student as CONTRIBUTOR [Version : 1]		
Author:	rawa	
Summary: Owner invites same-faculty student as CONTRIBUTOR. **Test design technique:** EP (role = CONTRIBUTOR partition) **Requirements:** REQ-SHR-03, REQ-SHR-07		
Preconditions: Student A owns SECJ3343. Student B has no invitation for this course.		
#:	Step actions:	Expected Results:
1	On the Collaborators page enter Student B's email, select 'Contributor', click Invite.	Invitation created with role Contributor, status Pending.
2	Log in as Student B, open Settings > Pending Invitations.	Invitation for SECJ3343 is listed showing permission Contributor and owner Student A.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-SHR-03: Invitation carries a permission level [Version : 1] REQ-SHR-07: Owner manages collaborators [Version : 1]	
Keywords:	None	

Test Case US-3: Non-owner cannot invite collaborators [Version : 1]		
Author:	rawa	
Summary: Non-owner cannot invite collaborators. **Test design technique:** DT (actor = collaborator/other student), negative **Requirements:** REQ-SHR-01		
Preconditions: Student B is an ACCEPTED CONTRIBUTOR of SECJ3343 (owner Student A). Student C exists.		
#:	Step actions:	Expected Results:
1	Log in as Student B, open SECJ3343 > Collaborators.	Invite controls are hidden or disabled for a non-owner.
2	As Student B send an invite request for Student C to SECJ3343 (e.g. via the API course.inviteCollaborator).	Request rejected with FORBIDDEN 'Only the owner can invite collaborators'; no invitation created.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-SHR-01: Only the course owner can invite collaborators [Version : 1]	
Keywords:	None	

Test Case US-4: Owner cannot invite himself [Version : 1]	
Author:	rawa

Summary: Owner cannot invite himself. **Test design technique:** EG / EP invalid class (invitee = owner) **Requirements:** REQ-SHR-02		
Preconditions: Student A logged in, owns SECJ3343.		
#:	Step actions:	Expected Results:
1	On the Collaborators page enter Student A's own email, choose Viewer, click Invite.	Error 'You can't invite yourself'; no collaborator row added.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-SHR-02: Invalid invitations are rejected [Version : 1]	
Keywords:	None	

Test Case US-5: Duplicate invitation to the same student is rejected [Version : 1]		
Author:	rawa	
Summary: Duplicate invitation to the same student is rejected. **Test design technique:** EP invalid class (already invited), DT **Requirements:** REQ-SHR-02		
Preconditions: Student B already has a PENDING (repeat with ACCEPTED) invitation to SECJ3343.		
#:	Step actions:	Expected Results:
1	Invite Student B again to SECJ3343 as Contributor.	Error 'User already invited'; existing invitation unchanged (role and status not modified).
2	Repeat with Student B's invitation in ACCEPTED status.	Same error 'User already invited'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-SHR-02: Invalid invitations are rejected [Version : 1]	
Keywords:	None	

Test Case US-6: Invite to a course that does not exist [Version : 1]		
Author:	rawa	
Summary: Invite to a course that does not exist. **Test design technique:** EG, EP invalid course id **Requirements:** REQ-SHR-01		
Preconditions: Student A logged in.		

#:	Step actions:	Expected Results:
1	Call course.inviteCollaborator with courseId 'does-not-exist', a valid user id and role VIEWER.	NOT_FOUND 'Course not found'; nothing stored.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Low	
Requirements	REQ-SHR-01: Only the course owner can invite collaborators [Version : 1]	
Keywords:	None	

Test Case US-7: Invitation to a student from a different faculty is blocked [Version : 1]		
Author:	rawa	
Summary: Invitation to a student from a different faculty is blocked. **Test design technique:** EP (same faculty / different faculty partitions) **Requirements:** REQ-SHR-08		
Preconditions: Student A (Faculty of Computing) owns SECJ3343. Student C is in Faculty of Engineering.		
#:	Step actions:	Expected Results:
1	Search/enter Student C's email on the Collaborators page and click Invite as Viewer.	System refuses the invitation with a message that only students of the same faculty can be invited; no invitation or notification is created.
2	Invite Student B (same faculty) the same way.	Invitation created (valid partition) - confirms the rule is faculty based.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-SHR-08: Faculty-restricted sharing [Version : 1]	
Keywords:	None	

Test Case US-8: Owner changes collaborator permission and removes collaborator [Version : 1]		
Author:	rawa	
Summary: Owner changes collaborator permission and removes collaborator. **Test design technique:** ST (VIEWER -> CONTRIBUTOR -> removed) **Requirements:** REQ-SHR-07		
Preconditions: Student B is ACCEPTED VIEWER on SECJ3343.		
#:	Step actions:	Expected Results:
1	As Student A open Collaborators, change Student B's role to Contributor and save.	Student B shown as Contributor; Student B can now add a resource (see TS-R3).
2	Change Student B back to Viewer.	Role shown as Viewer; Student B can no longer add resources.

3	Click Remove for Student B and confirm.	Student B disappears from the list; SECJ3343 no longer appears in Student B's courses.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-SHR-07: Owner manages collaborators [Version : 1]	
Keywords:	None	

4.2.Test Suite : TS-R2 Respond to Course Invitation

Invitee accepts / declines. Techniques: ST (PENDING -> ACCEPTED/REJECTED), DT.

Test Case US-9: Invitee accepts a pending course invitation [Version : 1]		
Author:	rawa	
Summary: Invitee accepts a pending course invitation. **Test design technique:** ST (PENDING -> ACCEPTED), use case main flow **Requirements:** REQ-SHR-04, REQ-NOT-01		
Preconditions: Student B has a PENDING VIEWER invitation to SECJ3343 from Student A.		
#:	Step actions:	Expected Results:
1	Log in as Student B, open Settings > Pending Invitations.	Invitation listed with course title, owner name and permission Viewer.
2	Click Accept.	Invitation removed from Pending Invitations; success message shown.
3	Open Courses.	SECJ3343 Software Quality Assurance is listed as a shared course with role Viewer.
4	Log in as Student A and open the bell.	Notification 'Invitation accepted': 'Student B accepted your invitation to SECJ3343 Software Quality Assurance'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1] REQ-NOT-01: Event notifications [Version : 1]	
Keywords:	None	

Test Case US-10: Invitee declines a pending course invitation [Version : 1]	
Author:	rawa
Summary: Invitee declines a pending course invitation. **Test design technique:** ST (PENDING -> REJECTED) **Requirements:** REQ-SHR-04, REQ-NOT-01	

Preconditions: Student B has a PENDING invitation to SECJ3343.		
#:	Step actions:	Expected Results:
1	As Student B open Settings > Pending Invitations and click Decline.	Invitation removed from the list; when none remain 'No pending invitations' is shown.
2	Open Courses as Student B.	SECJ3343 is NOT listed.
3	As Student A open the bell.	Notification 'Invitation declined': 'Student B declined your invitation to SECJ3343 Software Quality Assurance'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1] REQ-NOT-01: Event notifications [Version : 1]	
Keywords:	None	

Test Case US-11: Answered invitation cannot be accepted or declined again [Version : 1]		
Author:	rawa	
Summary: Answered invitation cannot be accepted or declined again. **Test design technique:** ST (invalid transition from ACCEPTED/REJECTED), EG **Requirements:** REQ-SHR-04		
Preconditions: Student B's invitation to SECJ3343 is already ACCEPTED.		
#:	Step actions:	Expected Results:
1	Call course.acceptInvitation for SECJ3343 again as Student B.	Request fails (record not found for status PENDING); status stays ACCEPTED, joinedAt unchanged, no new notification to owner.
2	Call course.rejectInvitation for SECJ3343 as Student B.	Request fails; status stays ACCEPTED.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1]	
Keywords:	None	

4.3.Test Suite : TS-R3 Shared Course Permissions (Viewer / Contributor)

Decision table: role {Owner, Contributor, Viewer} x action {view, download, add, upload, delete resource, delete course}.

Test Case US-12: VIEWER can view and download but cannot modify [Version : 1]	
Author:	rawa
Summary: VIEWER can view and download but cannot modify. **Test design technique:** DT (role = VIEWER column)	

Requirements: REQ-SHR-05

Preconditions:

Student B is ACCEPTED VIEWER on SECJ3343; card 'Lecture Slides' holds week1.pdf.

#:	Step actions:	Expected Results:
1	Log in as Student B and open SECJ3343.	Course and all resource cards are displayed.
2	Download week1.pdf from 'Lecture Slides'.	File downloads successfully.
3	Look for Add resource / Upload / Edit / Delete controls.	Controls are not shown (read-only view).
4	Attempt resource.create on SECJ3343 via API as Student B.	FORBIDDEN 'Only course owner and contributors can create resources'.
5	Attempt to upload a file to 'Lecture Slides' via API as Student B.	FORBIDDEN 'You don't have permission to add files to this resource'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-SHR-05: VIEWER has read-only access [Version : 1]
Keywords:	None

Test Case US-13: CONTRIBUTOR can add resources and upload files [Version : 1]

Author:	rawa

Summary:

CONTRIBUTOR can add resources and upload files.

Test design technique: DT (role = CONTRIBUTOR, allowed actions)

Requirements: REQ-SHR-06

Preconditions:

Student B is ACCEPTED CONTRIBUTOR on SECJ3343.

#:	Step actions:	Expected Results:
1	As Student B open SECJ3343 and add resource card 'Tutorial 1'.	Card created and visible to Student A as well.
2	Upload tutorial1.pdf (2 MB) into 'Lecture Slides'.	File uploaded and listed in the card.
3	Edit the title of 'Tutorial 1' to 'Tutorial 1 - BVA'.	Change saved.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-SHR-06: CONTRIBUTOR can add but not delete [Version : 1]
Keywords:	None

Test Case US-14: CONTRIBUTOR cannot delete resources or the course [Version : 1]

Author:	rawa

Summary: CONTRIBUTOR cannot delete resources or the course. **Test design technique:** DT (role = CONTRIBUTOR, forbidden actions) **Requirements:** REQ-SHR-06		
Preconditions: Student B is ACCEPTED CONTRIBUTOR on SECJ3343.		
#:	Step actions:	Expected Results:
1	As Student B try to delete resource 'Lecture Slides'.	Delete control hidden; API call returns FORBIDDEN 'Only the course owner can delete resources'. Resource still exists.
2	As Student B open SECJ3343 > Settings and try 'Delete course'.	Not allowed; course still exists.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-SHR-06: CONTRIBUTOR can add but not delete [Version : 1]	
Keywords:	None	

Test Case US-15: Non-collaborator and declined user cannot open a private course [Version : 1]		
Author:	rawa	
Summary: Non-collaborator and declined user cannot open a private course. **Test design technique:** EP (no relation / REJECTED / PENDING partitions) **Requirements:** REQ-SHR-04, REQ-SHR-05		
Preconditions: Student C has no relation to SECJ3343; Student B's invitation is REJECTED.		
#:	Step actions:	Expected Results:
1	As Student C open /courses/<SECJ3343 id> directly.	Access denied / course not found; no resources shown.
2	As Student B (rejected) open the same URL.	Access denied.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1] REQ-SHR-05: VIEWER has read-only access [Version : 1]	
Keywords:	None	

4.4.Test Suite : TS-R4 Create & Edit Article (UC-S4-003)

My Articles > New Article / Edit. Techniques: BVA on title (1..200) and excerpt (0..500), EP on cover URL.

Test Case US-16: Create article saved as draft (valid data) [Version : 1]	
Author:	rawa
Summary: Create article saved as draft (valid data).	

Test design technique: Use case main flow, EP valid

Requirements: REQ-ART-02, REQ-ART-10

Preconditions:

Student A logged in.

#:	Step actions:	Expected Results:
1	Open My Articles > New Article.	Form shows Title*, Excerpt (Optional), Cover Image (Optional), Tags (Optional), Content.
2	Title 'Intro to Boundary Value Analysis', Excerpt 'Short guide to BVA', Tags 'Testing, QA', some content; click Save Draft.	Toast 'Article saved as draft'.
3	Open My Articles.	Article listed with status DRAFT, tags Testing and QA, read time >= 1 min.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-ART-02: New article starts as DRAFT [Version : 1] REQ-ART-10: My Articles [Version : 1]
Keywords:	None

Test Case US-17: Title length boundaries (0, 1, 200, 201 characters) [Version : 1]

Author:	rawa

Summary:

Title length boundaries (0, 1, 200, 201 characters).

Test design technique: BVA

Requirements: REQ-ART-01

Preconditions:

Student A on My Articles > New Article.

#:	Step actions:	Expected Results:
1	Leave Title empty (0 chars), click Save Draft.	Error 'Please enter a title for your article'; nothing saved.
2	Title = 'A' (1 char), Save Draft.	Saved as draft.
3	Title = 200 x 'a', Save Draft.	Saved as draft.
4	Title = 201 x 'a', Save Draft.	Rejected with validation error (max 200); nothing saved.
5	Title = 5 spaces only.	Treated as empty: 'Please enter a title for your article'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-ART-01: Article input validation [Version : 1]
Keywords:	None

Test Case US-18: Excerpt length boundaries (500 / 501 characters) [Version : 1]

Author:	rawa
Summary:	
Excerpt length boundaries (500 / 501 characters). **Test design technique:** BVA **Requirements:** REQ-ART-01	
Preconditions:	
Student A on New Article with valid title 'Excerpt test'.	

#:	Step actions:	Expected Results:
1	Leave Excerpt empty and save.	Saved (excerpt optional).
2	Excerpt = 499 characters, save.	Saved.
3	Excerpt = 500 characters, save.	Saved.
4	Excerpt = 501 characters, save.	Validation error (max 500); not saved.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-ART-01: Article input validation [Version : 1]
Keywords:	None

Test Case US-19: Cover image URL validation [Version : 1]	
Author:	rawa
Summary:	
Cover image URL validation. **Test design technique:** EP (valid URL / invalid text / empty) **Requirements:** REQ-ART-01	
Preconditions:	
Student A on New Article, title 'Cover test'.	

#:	Step actions:	Expected Results:
1	Cover image = https://utfs.io/f/cover.png, save.	Saved; cover shown on article card.
2	Cover image = 'not a url', save via API article.create.	Validation error (invalid url); not saved.
3	Cover image empty, save.	Saved without cover (optional).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-ART-01: Article input validation [Version : 1]
Keywords:	None

Test Case US-20: Duplicate titles get unique slugs [Version : 1]	
Author:	rawa

Summary:

Duplicate titles get unique slugs.

Test design technique: EG, EP (new slug / existing slug)

Requirements: REQ-ART-03

Preconditions:

No article with slug 'software-testing-101' exists.

#:	Step actions:	Expected Results:
1	Create article 'Software Testing 101!' and publish.	URL is /articles/software-testing-101 (punctuation removed, lower-case, dashes).
2	Create a second article with the same title and publish.	URL is /articles/software-testing-101-1.
3	Create a third one.	URL is /articles/software-testing-101-2.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-ART-03: Unique URL slug [Version : 1]
Keywords:	None

Test Case US-21: Author edits own article; tags are replaced [Version : 1]

Author:	rawa

Summary:

Author edits own article; tags are replaced.

Test design technique: Use case alternative flow

Requirements: REQ-ART-06, REQ-ART-03

Preconditions:

Student A has DRAFT 'Intro to Boundary Value Analysis' with tags Testing, QA.

#:	Step actions:	Expected Results:
1	Open My Articles > Edit; change title to 'BVA Explained', tags to 'Testing'; Save.	Toast 'Changes saved'.
2	Re-open the article.	Title 'BVA Explained', slug 'bva-explained', only tag Testing (QA removed).

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-ART-03: Unique URL slug [Version : 1] REQ-ART-06: Edit own article [Version : 1]
Keywords:	None

Test Case US-22: User cannot edit another user's article [Version : 1]

Author:	rawa

Summary:

User cannot edit another user's article.

Test design technique: DT (actor = non-author), negative

Requirements: REQ-ART-06

Preconditions:

Student A authored 'BVA Explained'. Student B logged in.

#:	Step actions:	Expected Results:
1	As Student B open /my-articles/<A's article id>/edit.	Edit not possible / redirected; no form for A's article.
2	Call article.update for that id with a new title as Student B.	FORBIDDEN 'You can only edit your own articles'; article unchanged.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-ART-06: Edit own article [Version : 1]
Keywords:	None

4.5.Test Suite : TS-R5 Article Lifecycle (Publish / Archive / Delete)

State transition: DRAFT -> PUBLISHED -> ARCHIVED, delete from any state.

Test Case US-23: Publish a draft article [Version : 1]

Author:	rawa

Summary:

Publish a draft article.

Test design technique: ST (DRAFT -> PUBLISHED)

Requirements: REQ-ART-04

Preconditions:

Student A has DRAFT 'BVA Explained'.

#:	Step actions:	Expected Results:
1	In My Articles click Publish on 'BVA Explained'.	Toast 'Article published!'; status PUBLISHED, published date = today.
2	Log out and open /articles.	'BVA Explained' appears first in the list.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-ART-04: Publish workflow [Version : 1]
Keywords:	None

Test Case US-24: Only DRAFT articles can be published [Version : 1]

Author:	rawa

Summary:

Only DRAFT articles can be published.

Test design technique: ST (invalid transitions PUBLISHED->PUBLISHED, ARCHIVED->PUBLISHED)

Requirements: REQ-ART-04

Preconditions: Student A has one PUBLISHED and one ARCHIVED article.		
#:	Step actions:	Expected Results:
1	Call article.publish on the PUBLISHED article.	BAD_REQUEST 'Only draft articles can be published'; publishedAt unchanged.
2	Call article.publish on the ARCHIVED article.	BAD_REQUEST 'Only draft articles can be published'; status stays ARCHIVED.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-ART-04: Publish workflow [Version : 1]	
Keywords:	None	

Test Case US-25: Archive own published article [Version : 1]		
Author:	rawa	
Summary: Archive own published article. **Test design technique:** ST (PUBLISHED -> ARCHIVED) **Requirements:** REQ-ART-05, REQ-ART-07		
Preconditions: 'BVA Explained' is PUBLISHED by Student A.		
#:	Step actions:	Expected Results:
1	Open the article as Student A and choose Archive.	Toast 'Article archived'; status ARCHIVED.
2	Open /articles as a visitor.	'BVA Explained' no longer listed.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-ART-05: Archive and delete own article [Version : 1] REQ-ART-07: Public article listing [Version : 1]	
Keywords:	None	

Test Case US-26: Delete own article [Version : 1]		
Author:	rawa	
Summary: Delete own article. **Test design technique:** ST (any -> deleted) **Requirements:** REQ-ART-05		
Preconditions: Student A has article 'Delete me' (DRAFT).		
#:	Step actions:	Expected Results:

1	Open the article and choose Delete, confirm.	Toast 'Article deleted successfully'; redirected; article gone from My Articles.
2	Open its old URL /articles/delete-me.	'Article not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-ART-05: Archive and delete own article [Version : 1]	
Keywords:	None	

Test Case US-27: Non-author cannot publish, archive or delete [Version : 1]		
Author:	rawa	
Summary: Non-author cannot publish, archive or delete. **Test design technique:** DT (actor x action), negative **Requirements:** REQ-ART-04, REQ-ART-05		
Preconditions: Student A authored DRAFT X and PUBLISHED Y. Student B logged in.		
#:	Step actions:	Expected Results:
1	As Student B call article.publish on X.	FORBIDDEN 'You can only publish your own articles'.
2	As Student B call article.archive on Y.	FORBIDDEN 'You can only archive your own articles'.
3	As Student B call article.delete on Y.	FORBIDDEN 'You can only delete your own articles'; Y still exists.
4	As Student B open Y's page.	No Edit / Archive / Delete buttons shown.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-ART-04: Publish workflow [Version : 1] REQ-ART-05: Archive and delete own article [Version : 1]	
Keywords:	None	

4.6.Test Suite : TS-R6 Browse & Read Public Articles

/articles and /articles/[slug]. Techniques: EP, BVA on paging, DT on access.

Test Case US-28: Access control by status and viewer [Version : 1]
Author: rawa
Summary: Access control by status and viewer. **Test design technique:** DT (status {PUBLISHED, DRAFT, ARCHIVED} x viewer {visitor, other student, author}) **Requirements:** REQ-ART-08
Preconditions: Student A has PUBLISHED P, DRAFT D, ARCHIVED R.

#:	Step actions:	Expected Results:
1	Logged out, open /articles/<P slug>.	Article P displayed.
2	Logged out, open /articles/<D slug> and /articles/<R slug>.	'This article is not published' for both.
3	As Student B open D and R.	'This article is not published'.
4	As Student A open D and R.	Both displayed with author controls.
5	Open /articles/no-such-slug.	'Article not found'.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-ART-08: Article access control [Version : 1]	
Keywords:	None	

Test Case US-29: Listing shows only published, newest first, with pagination [Version : 1]		
Author:	rawa	
Summary:		
Listing shows only published, newest first, with pagination. **Test design technique:** EP + BVA (13 articles vs page size 12) **Requirements:** REQ-ART-07		
Preconditions:		
Exactly 13 PUBLISHED articles (published on different times) and 2 DRAFT articles exist.		
#:	Step actions:	Expected Results:
1	Open /articles.	12 articles on page 1, newest first; no DRAFT articles.
2	Go to page 2.	Exactly 1 article (the oldest).
3	Call article.getAllPublished with limit 0, 1, 50, 51.	0 and 51 rejected by validation; 1 and 50 accepted.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-ART-07: Public article listing [Version : 1]	
Keywords:	None	

Test Case US-30: Search and tag filter [Version : 1]	
Author:	rawa
Summary:	
Search and tag filter. **Test design technique:** EP (title match / excerpt match / no match / case) **Requirements:** REQ-ART-07	

Preconditions:

Published 'BVA Explained' (excerpt 'boundary values', tag Testing) and 'Docker Basics' (tag DevOps).

#:	Step actions:	Expected Results:
1	Type 'bva' in 'Search articles by title, content, or tags...'.	Only 'BVA Explained' shown (case-insensitive title match).
2	Search 'BOUNDARY'.	'BVA Explained' shown (excerpt match).
3	Search 'zzzz'.	Empty result message.
4	Clear search, filter by tag Testing.	Only 'BVA Explained'.
5	Open the tag filter list.	Contains Testing and DevOps; tags used only by drafts are not listed.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-ART-07: Public article listing [Version : 1]
Keywords:	None

Test Case US-31: View counter increments only for published articles [Version : 1]

Author:	rawa

Summary:

View counter increments only for published articles.

Test design technique: EP (published / not published)

Requirements: REQ-ART-09

Preconditions:

'BVA Explained' PUBLISHED with 5 views; DRAFT D with 0 views.

#:	Step actions:	Expected Results:
1	Open 'BVA Explained' as a visitor.	View count becomes 6.
2	Refresh once.	View count becomes 7.
3	Author opens DRAFT D.	View count of D stays 0.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Low
Requirements	REQ-ART-09: View counter [Version : 1]
Keywords:	None

Test Case US-32: My Articles status filter [Version : 1]

Author:	rawa

Summary:

My Articles status filter.

Test design technique: EP (filter = all / DRAFT / PUBLISHED / ARCHIVED)

Requirements: REQ-ART-10

Preconditions:

Student A has 1 DRAFT, 2 PUBLISHED, 1 ARCHIVED; Student B has 1 PUBLISHED.

#:	Step actions:	Expected Results:
1	Open My Articles (all).	4 articles of Student A only, most recently updated first.
2	Filter Draft / Published / Archived.	1 / 2 / 1 articles respectively; Student B's article never shown.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Low
Requirements	REQ-ART-10: My Articles [Version : 1]
Keywords:	None

4.7.Test Suite : TS-R7 Notification Centre (UC-S4-004)

Bell icon in the header. Techniques: BVA on badge (0, 1, 9, 10), ST (unread -> read).

Test Case US-33: Unread badge boundaries (0, 1, 9, 10) **[Version : 1]**

Author:	rawa

Summary:

Unread badge boundaries (0, 1, 9, 10).

Test design technique: BVA

Requirements: REQ-NOT-02

Preconditions:

Student B logged in with 0 unread notifications.

#:	Step actions:	Expected Results:
1	Look at the bell.	No badge shown (0).
2	Generate 1 notification (Student A invites B to a course). Refresh.	Badge shows 1.
3	Generate notifications until B has 9 unread.	Badge shows 9.
4	Generate the 10th.	Badge shows '9+'.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-NOT-02: Unread badge [Version : 1]
Keywords:	None

Test Case US-34: Dropdown lists 20 most recent, newest first **[Version : 1]**

Author:	rawa

Summary:

Dropdown lists 20 most recent, newest first.

Test design technique: BVA (20 / 21 notifications), EP (empty)

Requirements: REQ-NOT-03

Preconditions:

Student B has 0 notifications.

#:	Step actions:	Expected Results:
1	Open the bell.	'No notifications yet'.
2	Create 21 notifications for B, open the bell.	Exactly 20 shown, newest at top; oldest (1st) not shown.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-NOT-03: Recent notifications dropdown [Version : 1]
Keywords:	None

Test Case US-35: Click invitation notification marks read and opens Pending Invitations [Version : 1]

Author:	rawa

Summary:

Click invitation notification marks read and opens Pending Invitations.

Test design technique: ST (unread -> read), use case

Requirements: REQ-NOT-04, REQ-SHR-04

Preconditions:

Student B has 3 unread, one of them 'New course invitation'.

#:	Step actions:	Expected Results:
1	Open the bell and click 'New course invitation'.	Navigates to /settings?tab=invitations showing the invitation.
2	Look at the bell.	Badge decreased from 3 to 2; the clicked item is shown as read.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	High
Requirements	REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1] REQ-NOT-04: Mark one as read [Version : 1]
Keywords:	None

Test Case US-36: Mark all as read [Version : 1]

Author:	rawa

Summary:

Mark all as read.

Test design technique: ST (all unread -> read)

Requirements: REQ-NOT-05

Preconditions:

Student B has 5 unread notifications.

#:	Step actions:	Expected Results:

1	Open the bell and click 'Mark all as read'.	All items shown as read; badge disappears.
2	Reload the page.	Still no badge (persisted).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-NOT-05: Mark all as read [Version : 1]	
Keywords:	None	

Test Case US-37: User cannot read or mark another user's notification [Version : 1]		
Author:	rawa	
Summary: User cannot read or mark another user's notification. **Test design technique:** EG, security negative **Requirements:** REQ-NOT-06		
Preconditions: Student A has notification N (id known). Student B logged in.		
#:	Step actions:	Expected Results:
1	As Student B open the bell.	N is not listed; unread count excludes A's notifications.
2	As Student B call notification.markAsRead with N's id.	Error 'Notification not found'; N stays unread for Student A.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	High	
Requirements	REQ-NOT-06: Notification privacy [Version : 1]	
Keywords:	None	

Test Case US-38: View all notifications page with paging [Version : 1]		
Author:	rawa	
Summary: View all notifications page with paging. **Test design technique:** EP + BVA (limit 0, 1, 100, 101) **Requirements:** REQ-NOT-09		
Preconditions: Student B has 25 notifications.		
#:	Step actions:	Expected Results:
1	Open the bell and click 'View all'.	Full notification page opens (/notifications) showing the first 20 and a way to load more.
2	Load more.	Remaining 5 shown; no duplicates.

3	Call notification.getAll with limit 0, 1, 100, 101.	0 and 101 rejected; 1 and 100 accepted.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-NOT-09: Full notification history [Version : 1]	
Keywords:	None	

4.8.Test Suite : TS-R8 Notification Preferences

Settings > Notification Preferences. Techniques: EP, BVA (5..30 minutes), DT (channel x type).

Test Case US-39: Default preferences are created on first visit [Version : 1]		
Author:	rawa	
Summary: Default preferences are created on first visit. **Test design technique:** EP (new user without preferences) **Requirements:** REQ-NOT-07		
Preconditions: Newly approved Student E who never opened Settings.		
#:	Step actions:	Expected Results:
1	Open Settings > Notification Preferences.	Page loads with default toggles; reminder timing shows 10 minutes.
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-NOT-07: Notification preferences [Version : 1]	
Keywords:	None	

Test Case US-40: Save in-app / email preferences per type [Version : 1]		
Author:	rawa	
Summary: Save in-app / email preferences per type. **Test design technique:** DT (2 channels x 4 types) **Requirements:** REQ-NOT-07		
Preconditions: Student B on Notification Preferences.		
#:	Step actions:	Expected Results:
1	Turn OFF email 'Collaboration invites' and in-app 'Audit log alerts', keep the rest ON; Save.	Success message.
2	Reload page.	Exactly those two toggles are OFF.
3	Student A invites Student B to a course.	In-app notification appears; no invitation email is sent to Student B.

Execution type:	Manual
Estimated exec. duration (min):	
Importance:	Medium
Requirements	REQ-NOT-07: Notification preferences [Version : 1]
Keywords:	None

Test Case US-41: Class reminder timing boundaries [Version : 1]		
Author:	rawa	
Summary: Class reminder timing boundaries. **Test design technique:** BVA (4, 5, 30, 31 minutes) **Requirements:** REQ-NOT-08		
Preconditions: Student B on Notification Preferences.		
#:	Step actions:	Expected Results:
1	Open 'Reminder Timing' dropdown.	Options 5, 10, 15, 30 minutes only.
2	Select 5 minutes, save, reload.	5 minutes persisted.
3	Select 30 minutes, save, reload.	30 minutes persisted.
4	Call notification.updatePreferences with classReminderMinutes 4 and 31.	Both rejected by validation; stored value unchanged (30).
Execution type:	Manual	
Estimated exec. duration (min):		
Importance:	Medium	
Requirements	REQ-NOT-08: Reminder timing range [Version : 1]	
Keywords:	None	